Prompted to Start: How Generative AI is Transforming Entrepreneurship*

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Abstract

We show that following the diffusion of GenAI, industries with higher task-level exposure experience 20% more startup formation than less-exposed industries, and a growing share of these startups offer AI products rather than merely automating operations. Entry extends beyond innovation hubs into regions with limited venture capital and thin specialized labor markets. While individual entrants are smaller, aggregate entry generates net employment and wage growth at the industry level, including in the occupations most exposed to GenAI. GenAI also changes who becomes an entrepreneur: founders are more specialized, come from high-GenAI-exposure occupations, and emphasize technical skills rather than capital-raising expertise.

Keywords: *AI, Entrepreneurship, Startups, Founder, Agglomeration*

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1 Introduction

Generative artificial intelligence (GenAI) is increasingly viewed as a general-purpose technology that will reshape production and innovation through its broad applicability.[1] Accordingly, some observers emphasize the potential for substantial productivity gains and faster economic growth (Jones, 2024, Brynjolfsson, Li, and Raymond, 2025). At the same time, critics argue that GenAI may automate cognitive tasks without generating commensurate productivity gains, raising concerns about labor displacement (Acemoglu, Autor, Hazell, and Restrepo, 2022, Felten, Raj, and Seamans, 2023). Existing evidence remains inconclusive, in part because the adoption and diffusion of new technologies are gradual. We contribute to this debate by examining how GenAI affects entrepreneurial entry and the characteristics of new ventures. Entry decisions are inherently forward-looking. Therefore, the effects of a new general-purpose technology should surface earlier in new ventures than in macroeconomic outcomes or incumbent firms, which typically adopt new technologies more slowly (Crouzet, He, Lyonnet, and Ma, 2026). Because startups play a central role in job creation and reallocation (Haltiwanger, Jarmin, and Miranda, 2013, Decker, Haltiwanger, Jarmin, and Miranda, 2016, Adelino, Ma, and Robinson, 2017), their organizational choices offer a leading indicator of broader economic impacts.

We conjecture that GenAI is not simply a cost-reducing technology but may spur entry by altering the characteristics of founders and the organization of new ventures. Traditionally, financially constrained founders had to perform many tasks, and only individuals with balanced skills preferred to become entrepreneurs, especially when financial constraints prevented hiring complementary team members (Lazear, 2004, 2005). GenAI allows founders to automate some tasks and augment individual productivity in others at a per-unit usage fee that is negligible compared to wages.[2] As we show with the help of a stylized model, this allows more specialized individuals without large

[1] For classic discussions of general-purpose technologies and their implications for long-run growth, see Bresnahan and Trajtenberg (1995) and Jovanovic and Rousseau (2005).

[2] OpenAI's own estimates suggest that frontier models can complete tasks roughly 100x faster and 100x cheaper than industry experts. See `https://openai.com/index/gdpval/`.

financial resources to start their own businesses and create jobs in the very occupations whose productivity is augmented by GenAI, providing a countervailing mechanism to job displacements at incumbents. Entrepreneurial entry is further reinforced by the fact that the new technology may favor the creation of new products, increasing the return to entrepreneurship. We explore the implications of this mechanism for the characteristics of new ventures, their labor demand, and their founders' traits.

Our empirical strategy to capture the effects of GenAI on entrepreneurship exploits variation in industries' exposure to GenAI based on the task content of occupations. We measure task-level exposure using the actual proportion of queries to GenAI systems for specific tasks, reflecting how tech-savvy early adopters use the technology (Handa, Tamkin, McCain, Huang, Durmus, Heck, Mueller, Hong, Ritchie, Belonax, et al., 2025). Following Autor, Levy, and Murnane (2003), we aggregate task exposure to occupation and industry levels, yielding time-invariant measures that predict differential treatment intensity as GenAI diffuses. We then employ difference-in-differences specifications comparing entrepreneurial outcomes in high- versus low-exposure industries before and after GenAI's public release, beginning with Stable Diffusion in August 2022 and accelerating after ChatGPT's launch in November 2022.

We start by documenting that GenAI is indeed affecting entrepreneurial entry. Industries with high GenAI exposure experience a 20% increase in startup formation relative to low-exposure industries following ChatGPT's release. This effect is statistically significant starting in 2022Q4 and grows over time, indicating that GenAI induces substantial new entry rather than merely shifting the timing of marginal ventures. Using complementary LLM-assisted pipelines, we provide direct evidence that startups in industries exposed to GenAI are adopting the new technology and highlight two channels through which GenAI favors entrepreneurial entry: "AI-enabled" startups automate and augment existing business processes using GenAI, and "AI-native" ventures offer new products built on GenAI capabilities. The presence of AI-native startups suggests that GenAI is creating new products rather than simply reducing entry costs and may thus lead to exponential growth (Aghion and Howitt, 1992, Klette and Kortum, 2004, Jones, 2024).

Having established that GenAI is affecting entrepreneurial entry, we test the empirical implications of our conceptual framework. By automating tasks and augmenting skills, GenAI should decrease demand for highly specialized labor and external finance. Consistent with this prediction, we show that GenAI-enabled entry is geographically dispersed beyond traditional innovation hubs. New startups in high-GenAI-exposure industries increasingly locate in regions with limited venture capital access and thin specialized labor markets. Furthermore, startups in high-exposure industries enter at a smaller scale and offer a higher proportion of occupations that can be augmented by GenAI. Higher entry generates net employment growth and more than offsets smaller firm size, resulting in a 7% increase in employment and a 5% increase in average earnings in new firms in high-exposure industries.

Our stylized model also implies that GenAI should fundamentally alter the composition of founders, favoring the entry of specialized individuals. We find that following the diffusion of GenAI, founders in high-exposure industries are significantly more likely to have prior work experience in the industry of the startup and to have held GenAI-exposed occupations, enabling them to combine domain expertise with the productivity gains the new technology offers. Consistent with this interpretation, founders' self-described capabilities emphasize technical skills over capital-raising and deal-making skills. Both the shift in founder skills and the increase in entry outside venture capital hubs are consistent with GenAI reducing new firms' dependence on external finance. Furthermore, founders come from companies that were expanding employment, have compensation comparable to founders of startups less exposed to GenAI, and do not appear to experience compensation cuts in their transition to entrepreneurship. These findings suggest that founders are not pushed into entrepreneurship by uncertain job prospects at their previous employers and that they are not lower quality than pre-GenAI entrepreneurs. This supports our hypothesis that GenAI is not simply a cost-reducing technology which allows the creation of lower quality businesses. This interpretation is further corroborated by our finding that AI-enabled and AI-native startups are less likely to fail than other startups, as proxied by their websites going offline.

While the diffusion of GenAI coincides with the recovery from the COVID-19 pandemic, we

show that our results are unlikely to depend on firm dynamics associated with the pandemic, industry exposure to work from home, or a specific measure of industry exposure to GenAI.

Overall, our results suggest that business dynamism can mediate the labor-demand contractions GenAI produces among incumbents and that entrepreneurial entry constitutes a previously neglected countervailing channel. Aggregate entry more than offsets startups' smaller scale, generating net employment and wage growth in new firms at the industry level. Furthermore, while established companies demand fewer workers in GenAI-exposed occupations (Acemoglu, Autor, Hazell, and Restrepo, 2022), startups hire precisely these workers to exploit the new opportunities the technology creates.

Our paper contributes to several strands of the literature. A growing body of work examines GenAI's labor-market implications. Theoretically, the implications for the demand for specific skills and wage inequality are ambiguous (Agrawal, McHale, and Oettl, 2026). Acemoglu, Autor, Hazell, and Restrepo (2022) and Brynjolfsson, Chandar, and Chen (2025) document increasing demand for AI skills and a decrease in non-AI hiring and GenAI-exposed occupations for established firms. These changes appear to be accompanied by productivity gains for less-exposed workers (Hampole, Papanikolaou, Schmidt, and Seegmiller, 2025). We document that founders emphasize their AI skills also in startups; however, in startups, AI skills appear to be associated with job creation in GenAI-exposed occupations, possibly contributing to the redesign of these jobs towards less GenAI-automatable tasks (Brynjolfsson, Mitchell, and Rock, 2018). In contemporaneous work, Gupta, Qian, Simintzi, and Sun (2026) study how GenAI reshapes employment, productivity, and financing within existing high-growth startups, documenting a 3% workforce contraction at exposed firms alongside accelerated scaling and fundraising. Our focus is different: we examine the extensive margin of entry and the reallocation of human capital toward new firm formation. Not only do we estimate a much larger 25% drop in the number of employees in GenAI-exposed startups entering following the GenAI diffusion, but we also show that GenAI changes founding teams' characteristics.[3]

[3]The growth-oriented (transformational) startups in Crunchbase are not representative of the average new firm. So,

The literature on how GenAI shapes entrepreneurial entry is still in its early stages. Gofman and Jin (2024) show that the departure of AI professors reduces the probability that university alumni start an AI business. Their evidence is consistent with our finding that technical skills are important for AI entrepreneurs. Bao, Lou, and Sun (2025) focus on the tech sector, where GenAI may have pushed displaced individuals into entrepreneurship. Contemporaneous work by Marchesi and Tang (2026) documents industry-level entry consistent with reduced experimentation costs, and Cai, Gu, Sheng, Xia, Zhao, and Zhu (2025) report a post-ChatGPT rise in small-firm registrations in roughly 5 km^2 grid cells in China that had pre-2020 AI patents. We use a more general measure of exposure based on tasks actually performed through GenAI, document the mechanisms of technological diffusion that drive entry by measuring adoption directly, and assess the consequences of GenAI adoption for the internal organization of startups and the characteristics of founding teams.

Another strand of literature studies firm-level GenAI exposure in public companies. Eisfeldt, Schubert, Zhang, and Taska (forthcoming) find that high-exposure firms earned excess returns attributable to technology-labor substitution following ChatGPT's release. Babina, Fedyk, He, and Hodson (2024) and Babina, He, and Jiang (2026) document positive effects of AI workforces on listed firms' growth, innovation, and early productivity gains, while field experiments show that GenAI improves worker productivity within firms (Brynjolfsson, Li, and Raymond, 2025). However, macro-level impacts on productivity and employment remain negligible (Bick, Blandin, and Deming, 2025, Humlum and Vestergaard, 2026). By focusing on entrepreneurial ventures, we capture early-stage economic effects, and our findings suggest that GenAI may eventually drive productivity growth through startup formation and business model innovation.

Finally, we contribute to research on how technological change affects industrial structure. Previous work shows that data and computing power favor concentration and incumbent advantages (Eeckhout and Veldkamp, 2022, Lu, Phillips, and Yang, 2024), though cloud computing reduces capital requirements for AI startups (Ewens, Nanda, and Rhodes-Kropf, 2018). We show that

the large drop in firm size we estimate is consistent with an increase in employment in startups in the aggregate statistics we estimate, even if the increase in entry is only 20%.

GenAI tools promote entry and that, unlike cloud computing, they are not merely cost-reducing technologies but also favor the entry of specialized individuals who would previously have preferred paid employment.

2 Conceptual Framework

We develop a stylized model to guide our empirical analysis of how GenAI affects entrepreneurial entry and labor demand at startups. Unlike previous cost-reducing technologies, which required large upfront investments in fixed capital that only large incumbents could afford (Lashkari, Bauer, and Boussard, 2024, Hubmer and Restrepo, 2026), GenAI is priced per unit of use, at fees that are negligible relative to wages. It can automate some support activities and augment the productivity of individuals performing some core activities. Through these channels, GenAI may alter the skills of founders and the occupations created by startups and relax the financing requirements for entry.

2.1 Setup

Our framework follows Lazear (2005). A venture requires indispensable tasks. The *core* tasks embody the founder's idea or product, while the *support* task bundles activities such as marketing, administration, fundraising, and coordination. Potential entrepreneur i is characterized by a vector of skills $(x_{i1}, x_{i2}, s_i) \in (0,1]^3$: two core skills, ordered so that she is always better in the first, $x_{i1} \geq x_{i2}$, together with a support skill s_i.

Entrepreneurship is an occupational choice. As an employee, an individual specializes in the task in which she earns the most. For simplicity, all skills have the same market price w, so her wage is $w \max\{x_{i1}, s_i\}$: employment rewards an individual's best skill and is indifferent to the others. As an entrepreneur, the individual implements an idea which in expectation earns λQ_i, where Q_i is the venture quality and λ is the return to the venture quality. Tasks are indispensable, so venture quality is the weakest link across them. This weakest-link technology captures the idea that a strong product

cannot generate revenues without adequate execution, while excellent support cannot compensate for a weak product. Starting a venture also requires a fixed cost $c > 0$.

Each task can be performed in one of three modes. *Making* a task costs the founder nothing, but she can make a task only if she is proficient enough to reach the quality the venture targets. *Buying* a task means hiring, and the wage is paid per unit of the skill engaged: hires with combined skill h cost wh. Because the wage bill is linear in skill, only the combined skill devoted to a task matters, not how many people supply it. *Botting* is available on the support task alone: GenAI performs it outright at a metered fee that is negligible relative to wages, and which we therefore set to zero. While botting substitutes labor, GenAI can also *augment* the second core task, so that one or more persons of combined skills x working with a tool of quality z deliver $\psi(x,z)$, increasing in both arguments, with $\psi(x,0) = x$; the technology augments the founder or the hires performing that task. Task 1 embodies the founder's idea, is not augmented, and is always performed by her. Whenever the founder is indifferent between two modes, she makes.

Potential entrepreneurs also differ in their access to funds. Individual i is located in region ℓ, holds wealth $a_i \geq 0$, and can raise at most κ_ℓ from outside investors, so the resources she can commit to a venture are $A_i \equiv a_i + \kappa_\ell$. We interpret κ_ℓ as the depth of local early-stage finance, which is larger in venture capital hubs. Expenses are paid before revenue accrues, so a venture can be started only if its fixed cost and its wage bill can be funded out of A_i. The metered fee for botting is negligible and requires no funding.

2.2 Entry and the Founder Pool

We first isolate the margin that determines who becomes a founder. We consider an individual who cannot hire and has no access to outside funds, so that the relevant choice is between making and botting, and we accordingly collapse the two core tasks into one, $x_{i1} = x_{i2} \equiv x_i$. Section 2.3 lets the venture hire and raise outside funds, and shows that the shift in the founder pool derived here is reinforced.

Before GenAI, venture quality is $Q_i^0 = \min\{x_i, s_i\}$ and individual i enters if $\lambda \min\{x_i, s_i\} - c \geq w \max\{x_i, s_i\}$. This condition selects balanced individuals into entrepreneurship. To see this, consider an individual whose comparative advantage is in the core task, $x_i \geq s_i$. Her wage is wx_i, but venture quality is s_i. She enters only if $s_i \geq (wx_i + c)/\lambda$: the stronger her core skill, the more she can earn in paid employment and, therefore, the stronger her support skill must be to induce entry. The analogous condition applies to support-dominant individuals. Entrepreneurs thus occupy a cone around the 45° line in (x_i, s_i) space.

Once the technology is available, the founder chooses between making and botting support. A core specialist, $x_i \geq s_i$, bots exactly the minimum level that prevents support from constraining venture quality, x_i, so that $Q_i^{\text{AI}} = x_i$. She therefore expects $\lambda x_i - c$ and enters if

$$x_i \;\geq\; \widehat{x}_i \equiv \frac{c}{\lambda - w}, \qquad \lambda > w. \tag{1}$$

Because the founder can always go on making both tasks, GenAI weakly expands the entry set. Unlike the pre-GenAI entry condition, (1) places no bound on the founder's support skill. The founder's surplus over the wage she gives up, $(\lambda - w)x_i$, grows with her core skill and must cover the fixed cost of starting up. GenAI consequently adds founders with sufficiently strong core skills who would previously have remained employees because their support skills were too weak, and the founder pool shifts from balanced generalists toward core specialists. Figure A.1 provides a graphical representation of these effects. Because a solo founder incurs no wage bill, this comparison understates the cost of entry before GenAI, when a venture that cannot bot support must pay for it.

Comparative statics on the entry condition also highlight two complementary channels through which GenAI favors entry. *AI-enabled* startups are brought within reach of founders by automating and augmenting some tasks; the model captures this channel holding λ fixed. If the underlying set of opportunities is fixed, the resulting increase in entry may itself erode the return to quality. Although we do not model that equilibrium effect, a mere increase in GenAI-enabled startups is unlikely to lead to sustained growth and large increases in aggregate output. GenAI can also generate ideas

for new products. These *AI-native* startups can be viewed as having higher λ, the return to venture quality. A higher return to quality lowers the entry threshold (1), so entry extends to founders with lower core skills.

2.3 Staffing and Financing

Conditional on entry, we now let the two core tasks differ, $x_{i1} > x_{i2}$, and let the venture hire. We interpret $\psi(x,z)$ as the augmented task's quality. A target $\bar{q}$ is feasible only if $x_{i1} \geq \bar{q}$, and we assume that net operating surplus is strictly increasing in $\bar{q}$ over the feasible interval $[0, x_{i1}]$. The entrepreneur consequently chooses $\bar{q}_i^* = x_{i1}$: the founder's skill in task 1 determines the venture's endogenous target output. She can make task 2 if $\psi(x_{i2}, z) \geq \bar{q}$ and support if $s_i \geq \bar{q}$.

Because hiring commands a strictly positive wage while making costs nothing, the founder performs every task she can reach and hires only those she cannot. Support is therefore never bought: the founder covers it herself when $s_i \geq \bar{q}$ and bots it otherwise. Task 2 is made when $\psi(x_{i2}, z) \geq \bar{q}$, and bought otherwise at the wage bill $w\,\psi^{-1}(\bar{q}; z)$ of the smallest adequate team, where $\psi^{-1}(\cdot\,; z)$ inverts ψ in its first argument. Since $\psi(x,z) \geq x$, the combined skill the venture must buy is lower, and hence cheaper, than the skill $\bar{q}$ an unassisted task would require.

Setting $\bar{q} = x_{i1}$ describes the venture's staffing as a function of the founder's core skill alone. Before GenAI, the technology is absent, $z = 0$, so every venture hires for task 2 at a wage bill wx_{i1}, and ventures whose founder has $s_i < x_{i1}$ hire for support as well. After GenAI, support leaves the payroll at every venture, and task 2 stays on it only for founders whose core skill exceeds what assistance lets them deliver in the second task,

$$x_{i1} \;>\; \psi(x_{i2}, z). \tag{2}$$

For these founders, the wage bill falls from wx_{i1} to $w\,\psi^{-1}(x_{i1}; z)$, because augmentation lets less skilled and therefore cheaper hires clear the same target; for the rest it falls to zero. Both margins move with the quality of the technology: as z rises, more founders cover task 2 themselves, and

those who continue to hire do so more cheaply.

Staffing determines financing. If expenses must be paid before revenue, the venture's external funding requirement is its fixed cost plus its wage bill, $I_i = c + w\sum_{k\in B_i} h_k$, where B_i is the set of tasks the founder buys and h_k the combined skill she buys for task k: her own work requires no cash, and the metered bill is negligible. Since the wage bill falls at every venture, post-GenAI entrants launch with less outside funding and smaller teams, and depend less on outside investors.

Reinstating hiring and outside finance also sharpens the entry margin of Section 2.2. Individual i enters only if the venture is worth starting and she can fund it,

$$\lambda x_{i1} - c - W_i \;\geq\; w\max\{x_{i1}, s_i\}, \tag{3}$$

$$c + W_i \;\leq\; A_i, \tag{4}$$

where $W_i = w\sum_{k\in B_i} h_k$ is the venture's wage bill. Before GenAI, a founder whose support skill falls short of her target pays two wage bills, and (3) requires $x_{i1} \geq c/(\lambda - 3w)$, a stronger core skill than the threshold $\widehat{x}_i$ in (1). Shutting down hiring in Section 2.2 is therefore conservative: the shift of the founder pool toward core specialists is larger than Figure A.1 suggests.

The financing condition adds a margin that operates across locations. Because the wage bill scales with the target that the founder's own core skill sets, (4) caps entry at $x_{i1} \leq (A_i - c)/(2w)$ before GenAI and imposes no cap after if support and task 2 leave the payroll. No core specialist enters before GenAI whenever

$$c \;\leq\; A_i \;<\; c + \frac{2wc}{\lambda - 3w}, \tag{5}$$

while entry with sufficiently strong core skills becomes feasible afterward. Since $A_i = a_i + \kappa_\ell$, the more individuals fall in this window, the thinner is the local early-stage finance.

2.4 Empirical Implications

The framework delivers three sets of implications, which also help distinguish the effects of GenAI from an alternative in which investment opportunities brought by new technologies reduce entry costs and draw uniformly lower-quality individuals into entrepreneurship.

1. *Where entry occurs.* The founders whose entry GenAI makes possible are those who were constrained by their own skills and could not fund the hires that would have compensated for them. Since GenAI reduces new venture demand for wage funding and specialized labor, we expect an increase in entry outside venture capital hubs and specialized labor markets.

2. *The size and composition of entrants.* The startups whose entry is favored by GenAI are smaller, because the founder can achieve the same quality with fewer hires. Since founders do not hire in occupations that can be automated, a larger proportion of occupations they create involve tasks the technology augments. While startups are smaller, the increase in their number, together with the fact that the more specialized founders GenAI brings in are more likely to hire for tasks that are augmented, may involve an increase in aggregate employment at startups.

3. *Who becomes a founder.* Since the founder's own skills in the tasks that can be automated or augmented no longer constrain venture quality, the founders of startups whose entry is favored by GenAI should be more likely to be specialized in core technical skills related to the startups' activities and to have gained experience in the industry of the startup. Proxies for an individual's best skill, such as prior compensation, are not necessarily expected to decrease, precisely because GenAI favors the entry of specialists, not of less skilled individuals.

Below, we assemble a variety of datasets and test these implications for the effects of GenAI on entrepreneurial entry and the changing characteristics of startups and their founders.

3 Data and Measurement

3.1 Entrepreneurial Entry and Firm Dynamics

We begin with the U.S. Census Bureau's Business Formation Statistics (BFS), which is constructed by the Census Bureau using administrative data from the Internal Revenue Service (IRS) on Employer Identification Number (EIN) applications. All employer businesses in the United States are required to have an EIN to file payroll taxes. New nonemployer businesses also file for an EIN if forming a partnership or an incorporated business. Even new sole proprietors often file for an EIN to facilitate their business activities (e.g., working with other businesses or opening a business bank account). The Census Bureau publishes monthly the number of EIN applications submitted to the IRS and projected business formation statistics, computed using predictive models linking information from the EIN applications to subsequent business openings. Data are published monthly at the two-digit NAICS level and serve as leading indicators of nascent entrepreneurship across industries (Dinlersoz, Dunne, Haltiwanger, and Penciakova, 2021).

This dataset allows us to track the dynamics of business formation at high frequency, but at the cost of a rather coarse industry aggregation. To capture industry dynamics with higher granularity, we use *Data Axle* (previously named InfoGroup), which tracks the universe of U.S. establishments and contains information on founding year, location, industry classification (four-digit NAICS), and employment. The database is constructed from multiple administrative and commercial sources, including business directories, state registration records, telephone and utility connections, and verified survey data, and is widely used in academic research (Acharya, Bhardwaj, and Tomunen, 2023, Borisov, Ellul, and Sevilir, 2021, Denes, Howell, Mezzanotti, Wang, and Xu, 2023). InfoGroup's granularity enables analysis of new firm creation and size distribution at fine levels of industry aggregation.

Table 1 reports summary statistics for the main variables. Panel A presents two-digit industry-year-month business formation statistics, while Panel B presents the four-digit industry-year dataset

constructed from InfoGroup for the period 2020 to 2024. At the four-digit level, while almost 80% of industry-years experience entry, the number of firms entering varies widely across industries.

To assess the broader aggregate labor market implications of changes in entrepreneurial entry, we complement the above data sources with the Census Quarterly Workforce Indicators (QWI). QWI provides a nationally consistent measure of employment and average monthly earnings of employees in firms of different ages at the industry (four-digit NAICS) and quarter level. In our tests, we distinguish between very young firms (ages 0–1) and the rest of the companies to evaluate the employment effects of startups. Table B.1 provides summary statistics. Across industry-quarter cells, very young firms employ an average of 18,434 workers, with monthly earnings of $4,624.

3.2 Startups and Founding Teams' Characteristics

To study labor demand in new firms, we construct an employer-employee-matched database linking Crunchbase startups to LinkedIn employee data.

We first obtain data from Crunchbase, a database on innovative companies, used by the venture capital industry to identify new investments. We consider five annual snapshots of startups founded in the United States between 2020 and 2024. From Crunchbase, we obtain each startup's proprietary industry classification and the founding year.

We then link firm information from Crunchbase to employee-level data from LinkedIn, via Revelio Labs, using the firm-specific LinkedIn URLs provided in Crunchbase. We identify employees who list the startup as their current employer on LinkedIn as of the end of the founding year. Each employee's job title, reported at the eight-digit O*NET-SOC level, is mapped to a six-digit occupation code using the U.S. Department of Labor's O*NET classification system. Panel C of Table 1 summarizes the characteristics of all startups at the end of the founding year. On average, a startup has 4.6 employees spanning 3.3 distinct occupations by the end of its founding year.

We further identify the founders of each startup using the founders' records provided in Crunchbase. For each founder, we extract information on professional background and prior work experi-

ence from LinkedIn profiles, using the founder-specific LinkedIn URLs provided in Crunchbase. We can match LinkedIn profiles for founders of approximately one-third of the startups in the Crunchbase sample; Panel D of Table 1 summarizes founder characteristics.

4 A Forward-Looking Measure of GenAI Exposure

The extent to which startups can use GenAI depends on the nature of the tasks performed in their industry. Existing proxies for task-level GenAI exposure infer exposure from the similarity of job descriptions to textual descriptions of patents involving AI (Hampole, Papanikolaou, Schmidt, and Seegmiller, 2025), from expert assessments(Felten, Raj, and Seamans, 2021, Brynjolfsson, Mitchell, and Rock, 2018, Shao, Zope, Jiang, Pei, Nguyen, Brynjolfsson, and Yang, 2025), or from surveys of workers (Humlum and Vestergaard, 2026). These measures describe what AI can do against a snapshot of today's task organization.

Since our objective is to capture the effects of GenAI on startups that are early adopters of the new technology, we use the Anthropic Economic Index (AEI), which measures task reliance on GenAI from the realized choices of Claude.ai users. The AEI is constructed through a privacy-preserving analysis of millions of real-world conversations on Claude.ai. It measures the share of total queries involving a specific task among all queries to Claude.ai during a week, including API queries. The index, therefore, captures the extent to which a task is performed by GenAI. We use the February 2025 version of the AEI, which is based on conversations during December 2024.[4] Table B.2 provides examples of tasks with different exposure to GenAI. Because Claude.ai users are drawn disproportionately from early adopters of GenAI, the AEI captures the frontier of GenAI adoption and is well-suited to track the new technology's effects on early adopters.

Using this measure of task exposure to GenAI, we follow the methodology pioneered by Autor, Levy, and Murnane (2003) to obtain occupation and industry level proxies for GenAI exposure. Specifically, we first aggregate approximately 19,000 distinct task-level exposures into the 923

[4]For robustness, we use later releases, including the September 2025 release, which incorporates API usage data.

occupations defined by the U.S. Department of Labor's O*NET occupational classification system based on the fraction of each occupation's constituent tasks that can be automated or enhanced by GenAI. As expected, computer programmers and IT-related jobs appear to be highly affected. Table B.3 provides more examples of occupations with high, average, and low exposure to GenAI. Then, we map occupations to industries using the Bureau of Labor Statistics (BLS) Industry–Occupation Employment Matrix, which provides the employment share of each occupation within each industry. Formally, for each four-digit NAICS industry n, we compute:

$$AEI_n^{\text{NAICS4}} = \sum_o s_{on} \times AEI_o, \tag{6}$$

where AEI_o denotes the Anthropic Economic Index exposure score for occupation o, and s_{on} represents occupation o's share of total employment in industry n.

We compare our AEI_n^{NAICS4} index with two other widely used indices of AI exposure. Across the 183 industries for which all measures are available, the AEI_n^{NAICS4} rank correlation with Felten, Raj, and Seamans (2023) is 0.7 and with Eisfeldt, Schubert, Zhang, and Taska (forthcoming) and Eloundou, Manning, Mishkin, and Rock (2024) is 0.8, comparable to the rank correlation of 0.77 between the two benchmarks. The AEI_n^{NAICS4} therefore identifies broadly the same ordering of industries as these alternative measures. This reassures us that our results are not driven by idiosyncrasies of the Claude.ai user base or the time of the conversations.

Since our other data sources provide different industry classifications or levels of granularity, we further aggregate the four-digit NAICS industry level measure to the two-digit sector level by taking employment-share-weighted averages across four-digit industries within each sector. Moreover, Crunchbase uses a proprietary industry classification, and each startup is assigned to one or more category groups (e.g., "Artificial Intelligence," "Financial Services," "Healthcare"), which overlap and are not mutually exclusive.[5] We construct a measure of GenAI exposure based on the startups'

[5]While the merged dataset with Revelio includes a standard industry identifier (a six-digit NAICS code), this information is missing for roughly half of the firms in the combined Crunchbase–LinkedIn sample. Moreover, NAICS codes are not ideal for classifying Crunchbase startups because innovative startups are concentrated in a handful of NAICS industries. In contrast, Crunchbase's own industry group classification better captures emerging sectors and

pre-GenAI occupational composition, as observed in Revelio Labs data, within a Crunchbase industry category group. Specifically, we first calculate each startup's GenAI exposure for each year as the employment-weighted average of exposure across its occupations. For each Crunchbase category group, we then average these firm-level exposures across the startups assigned to that category and over the pre-GenAI years. Because a startup can belong to multiple category groups, we assign it the average pre-period exposure of all its assigned category groups. Effectively, each startup's exposure, denoted as AEI_f^{pre}, is determined by the pre-GenAI occupational composition of startups sharing its category affiliations.

Table B.4 provides examples of four-digit NAICS industries with high, median, and low levels of GenAI exposure. Software and other IT industries have high GenAI exposure; manufacturing, such as automobiles, and certain services, such as healthcare, have medium GenAI exposure; while industries that involve human interaction and manual work (e.g., delivery services, mining) have low GenAI exposure. Figure A.2 shows the distribution of GenAI exposure across the four-digit NAICS industries and the startups in our different samples. In Panel A, over 50% of industries appear to have very low exposure to GenAI. Exposure to GenAI varies significantly across the remaining industries. In the most exposed industries, over 10% of tasks can be performed using GenAI. The distribution of GenAI exposure is significantly higher in the merged Crunchbase-Revelio sample because firms in this sample are predominantly in the IT sector, as shown in Table B.5, and all of them have tasks that can be performed through GenAI. Table B.6 describes the top-ranked startups in terms of GenAI exposure. For instance, SuperDuperAi is an AI platform for creating professional videos. Experio Labs, Inc. offers an AI-powered knowledge management platform for startups.

Given the high dispersion of the GenAI industry exposure proxy, our tests compare high- and low-exposure industries, defined as those with exposure above and below the relevant sample median. We show in Section 5 that our results are robust to the measurement of task exposure to GenAI by using conversations with an alternative, widely used AI platform.

groups together firms engaged in comparable business activities.

5 GenAI and New Business Formation

Before testing the implications of our conceptual framework for how GenAI affects entrepreneurial entry and the characteristics of founding teams, we need to establish that entry is increasing and that our proxy for GenAI exposure can indeed capture GenAI adoption. To this end, we show that the diffusion of GenAI is associated with more entrepreneurial entry and provide direct evidence of how startups adopt the new technology.

5.1 Evidence on Entry

We use the BFS to test whether industries with greater GenAI task exposure experienced larger increases in entrepreneurial entry following its diffusion. We estimate the following difference-in-differences specification:

$$Y_{nmt} = \alpha + \beta \left(AEI_n^{\text{NAICS2}} \times Post_t \right) + \delta_n + \delta_{mt} + \varepsilon_{nmt}, \tag{7}$$

where the outcome variable, Y_{nmt}, is the number of projected new businesses in the two-digit NAICS industry n during month m of year t. We define the post-period indicator ($Post_t$) as equal to one from August 2022 onward, corresponding to the public release of *Stable Diffusion*, one of the first widely accessible text-to-image GenAI models. This timing coincides with the broader wave of mass-adopted foundation models, such as Midjourney's open beta in July 2022 and *ChatGPT*'s public release in November 2022, a period characterized by the improvement and rapid diffusion of foundation models and the widespread integration of generative tools into product development. The variable AEI_n^{NAICS2} denotes the two-digit NAICS industry-level AI exposure index, aggregated from occupation-level AEI scores. Estimates are obtained, including industry (δ_n) and month-year (δ_{mt}) fixed effects. The sample period ranges from January 2020 to May 2025. Given the small number of two-digit NAICS sectors, conventional cluster-robust inference at the sector level may be unreliable. We therefore report Driscoll–Kraay standard errors, allowing for both cross-sectional

dependence across sectors and serial correlation within sectors over four monthly lags.[6]

Our empirical model enables us to test whether GenAI is indeed favoring entry in industries in which a sufficient number of tasks are exposed to the technology. The coefficient β measures the differential change in projected business formation across industries with varying levels of pre-existing GenAI exposure following the diffusion of GenAI.

Table 2 reveals strong positive effects of GenAI diffusion on new business formation in industries relatively more exposed to GenAI. We consider different measures of new business formation, including: (i) All business applications (BA) (column 1); (ii) Applications from businesses that are estimated to have a high propensity to become employers, due to the presence of planned payroll in their applications (column 2), which we refer to as *High-Propensity Business Applications* (HBA); and (iii) Corporations' business applications (CBA) (column 3). While the effect on overall business applications is not statistically significant, industries with above-median GenAI exposure experience an 8.3% larger increase in monthly HBA, an effect that is both statistically significant and economically meaningful. The estimated effect is even larger for CBAs, which experience a 12% higher monthly increase following the diffusion of GenAI. The effects are also large for *Projected Business Formations* (PBF), the Census Bureau's projections of the applications that will become employers within four and eight quarters, which increase by 7.5% and 8.1% (columns 4 and 5). This evidence suggests that GenAI is not generating a proliferation of marginal ventures but favoring the entry of more formal businesses with the potential to become employers, as implied by our conceptual framework.

To address concerns that we may capture pre-existing trends, Figure 1 presents both the raw patterns and the corresponding dynamic estimates for CBAs. Panel A plots business formation separately for sectors with high and low GenAI exposure. The two groups evolve similarly before the diffusion of GenAI but begin to diverge in late 2022, with business formation increasing

[6]With four monthly lags, the kernel spans a full quarter of serial correlation; this is close to the Newey–West rule of thumb $4(T/100)^{2/9}$, which gives 3.6 at our sample length of $T = 65$ months. Estimates and inference are unchanged at six and twelve lags (Table B.7).

more strongly in high-exposure sectors thereafter. Panel B confirms this pattern in the regression framework: the estimated effects are close to zero before the diffusion of GenAI, become positive shortly thereafter, and increase over time as the technology spreads and improves. The absence of systematic differential pre-trends supports the validity of our difference-in-differences design.

The BFS data are well-suited to detecting the timing of entry, but two-digit NAICS industry coverage is coarse. We therefore turn to InfoGroup, which allows us to measure entry at the four-digit NAICS level. We consider newly established, single-location firms, treating each firm's founding year as the observation year. For each year between 2020 and 2024, we count newly founded firms in each four-digit NAICS industry. To capture the differential change in firm entry across industries with varying degrees of pre-existing GenAI exposure following the diffusion of such technologies, we estimate a difference-in-differences specification similar to regression (7) but with all industry variables defined at the four-digit level and at yearly frequency. Accordingly, the *Post* indicator takes a value equal to one for the years 2023 and 2024. We include industry as well as year fixed effects and cluster standard errors at the four-digit NAICS level.

Table 3 reports the estimates. We measure exposure to GenAI using each industry's quartile in the distribution of exposure or, alternatively, an indicator for above-median exposure. We find that in more AI-exposed industries, the probability of observing any new venture (an indicator equal to one if at least one new firm is established in an industry-year cell) increases after the diffusion of GenAI (columns 1 and 2). Column 2 of Table 3 implies that industries with GenAI exposure above the median have an 11-percentage-point higher probability of any new firm entering during a year following the diffusion of GenAI, or about 14% of the sample mean. This result suggests that the likelihood of new firm creation has increased in industries that were already technologically proximate to GenAI before 2023. Moreover, industries with greater pre-existing AI exposure have more new firms entering post-GenAI (columns 3 and 4 of Table 3). Results are again not only statistically but also economically significant. Column 4 of Table 3 implies that new firms increase by over 20% more in industries with GenAI exposure above the median following the introduction

of GenAI, relative to less-exposed industries.[7] Overall, it appears that GenAI lowered the fixed costs of entry or opened new market opportunities in these sectors.

Figures A.3 and A.4 show that these estimates are robust when considering different proxies for industries' exposure to GenAI, including one based on conversations on Microsoft's Bing Copilot, when excluding 2020 and controlling for industry exit rates during 2020-2021 to account for the possible impact of the COVID19 pandemic. Estimates are equally robust when we control for the industry's suitability for work from home, using data from Dingel and Neiman (2020). Overall, entrepreneurial entry appears to have increased in industries in which more tasks can be performed through GenAI. In the rest of this section, we explore how GenAI is affecting the operations and products of new businesses.

5.2 How Startups Use Generative AI

The patterns documented so far are consistent with GenAI encouraging entry, but they do not establish whether and how entrants actually use the technology. In this subsection, we show that a startup's industry exposure to GenAI is indeed associated with rapid GenAI adoption.

Startups may use GenAI to automate and augment their internal operations and to build new products. To capture these two mechanisms favoring entry, we distinguish between *AI-enabled* startups, defined as firms that use GenAI for internal operations, and *AI-native* startups, defined as firms whose products depend on AI. The two categories are not mutually exclusive: a firm that both sells AI products and uses AI internally is classified under both labels.

Classification methodology. Identifying AI-native and AI-enabled startups is challenging because Crunchbase profiles rarely document how a firm operates internally, and self-reported product labels are noisy. We therefore construct two complementary classification pipelines based on verifiable

[7] Table B.8 shows that the estimates are qualitatively invariant when we use count models. Since only 20% of the industry-year observations have zero entry, we present ordinary least squares estimates, which are easier to interpret, in the body of the paper. All two-digit NAICS industries have positive entry in Table 2.

external evidence. We describe both pipelines in detail in Appendix C.

The first pipeline identifies AI-native startups by combining Crunchbase information with evidence from company websites retrieved using Tavily. Because the product classification relies primarily on website text, restricting attention to currently retrievable websites would condition the sample on survival to the data collection date. For firms whose site no longer responds, we therefore recover an archived homepage from the Internet Archive's Wayback Machine, selecting a snapshot from approximately six months before the site's last available capture. Live and archived text are processed identically thereafter. Using archived websites raises coverage from 48% to 83.9% of the sample. This approach preserves coverage of firms that disappeared or rebranded.

The pipeline first determines whether AI or machine learning is essential to the firm's core product or service—that is, whether the offering would fundamentally exist without it. It then assigns each firm to a product taxonomy. AI-native categories comprise foundation-layer firms, AI infrastructure and tooling firms, thin LLM wrappers, thick LLM integrators, applied vertical AI firms, autonomous-agent systems, and generative-content platforms. Firms classified as not AI-native are assigned to traditional technology or software-as-a-service, AI-augmented, or non-technology categories. For each AI-native firm, the pipeline also measures Resource-Adjusted AI Dependency (RAD), which captures the firm's structural reliance on third-party foundation models and APIs. The measure distinguishes firms with high, medium, or low dependence on external models, allowing us to separate firms whose core products depend heavily on outside providers from those with substantial proprietary capabilities.

Since startup websites rarely describe how firms use GenAI in their internal operations, the second pipeline identifies AI-enabled startups through a three-stage agentic procedure. Stage 1 assesses whether a startup has sufficient public information to be researched reliably. It combines a website-status check with a query to the Tavily Search API and advances only firms with strong research signals. Stage 2 uses the Perplexity Agent API to search three evidence channels separately: job postings, firm-owned sources, and third-party sources. Short screening searches identify promising channels, after which more extensive searches collect citable evidence of internal GenAI

use. Stage 3 independently verifies each remaining claim by reopening its cited URL and comparing the retrieved text with the saved evidence summary.

The pipeline applies strict evidence standards. Every positive finding must be supported by a citable URL, and the source must explicitly describe the firm's internal use of GenAI rather than merely contain language from which adoption might be inferred. The pipeline also enforces a strict distinction between *using* and *selling*: evidence that a startup sells an AI product does not, by itself, establish that it uses GenAI in its own operations. We classify a startup as AI-enabled if it has at least one independently verified record documenting the use of a GenAI tool in an internal function, such as engineering, marketing, operations, customer service, sales, human resources, or finance.[8]

Importantly, for all AI-native and AI-enabled startups we identify, we check whether their websites are still live in 2026 to construct a proxy for firm survival. In the absence of financial information, this allows us to infer the quality of startups whose entry has been favored by GenAI.

Coverage. We begin with the 44,387 startups in the matched U.S. Crunchbase–Revelio sample. Of these, 9,420 (21.2%) pass the AI-enabled screening and advance to the evidence search, and 5,028 (11.3%) have at least one independently verified finding of internal GenAI use. The AI-native pipeline completes classification for 37,218 firms (83.9%) using current or archived website evidence and identifies 4,669 of them (12.5%) as AI-native. The two measures are distinct. Among the 8,252 startups examined by both pipelines, their correlation is 0.109.[9] Approximately two-fifths of AI-enabled startups are also AI-native, while three-fifths of AI-native startups are also AI-enabled.

Even though the pipelines to identify AI-native and AI-enabled firms are likely to understate

[8]For example, the pipeline classifies Sawmills as AI-enabled in engineering based on a company blog post in which its CTO describes using Claude Opus, GitHub Copilot, and GPT-based models for planning, execution, and debugging in software development. It classifies Infinite Renewals as AI-enabled in marketing based on a podcast in which the founder explicitly reports using Jasper for content creation. It classifies Daisy as AI-enabled in customer service based on a blog post explaining how technicians use ChatGPT to assist with troubleshooting and customer questions.

[9]We restrict this comparison to firms examined by both pipelines because firms that do not advance beyond the AI-enabled screen are mechanically coded as not AI-enabled.

adoption, it is comforting that in Figure 2, the shares of both AI-enabled and AI-native startups among founding cohorts have increased sharply since 2022. The presence of AI-enabled and AI-based startups before the fall of 2022 reflects the early applications of the technology, which were already available at that time. The subsequent rise is steeper for AI-native startups, consistent with a process of technological diffusion that has led to the emergence of new products built around generative capabilities. Thus, while GenAI is favoring entry by enabling startups to automate tasks and lowering entry costs, a substantial proportion of new firms are creating AI-based products, which could be conducive to exponential growth.

Adoption patterns. We explore whether AI-enabled and AI-native startups, for which we have direct evidence of adoption, can indeed explain the differences in entry rates we have documented so far. Figure A.5 plots the share of AI-enabled startups in each Crunchbase industry category group against the group's GenAI exposure, separately for each founding year before and after the new technology's diffusion.[10] The relation is positive in every year, but it becomes steeper in 2023 and 2024 relative to 2020 and 2021. Figure A.6 shows the same pattern for AI-native startups: the slope of the relation between the share of AI-native firms and category-level GenAI exposure steepens markedly after 2022, suggesting that technological diffusion has led many startups to develop AI products. This evidence corroborates our methodology based on industry exposure to GenAI.

Figure 3 shows the startups' internal business functions for which GenAI is used (panel (a)) and how the shares of mentioned use cases evolve across years (panel (b)). The most common uses are operations (26% of verified use cases), engineering (25%), and marketing (16%), followed by product and design, sales, customer service, human resources, and finance and accounting. All these functions involve tasks with high exposure to GenAI in our AEI measure, providing a further consistency check between the pipeline classification and the task-based exposure index used in our main specifications. Importantly, the shares associated with engineering and operations increase over time, while the share associated with marketing declines, suggesting a growing emphasis on

[10] We omit 2022, the intermediate year.

GenAI in startups' core production and operational activities.

The verified adoption evidence spans a broad set of GenAI tools: ChatGPT and other OpenAI products account for 49% of mentions, Claude and other Anthropic products for 22%, GitHub Copilot for 12%, Cursor for 11%, and Gemini and other Google AI products for 6%. The evidence is drawn from third-party sources such as news, interviews, and podcasts (38.0%), firm-owned media such as company websites and blogs (33.9%), and job postings (28.2%), so no single channel drives the measure. Importantly, the fact that AI-enabled startups name a broad range of tools reassures us that our AEI-exposure measure, though constructed from Claude.ai conversations, captures adoption patterns that extend well beyond any single platform.

Figure 4 describes the characteristics of AI-native startups. In panel (a), foundation-layer firms, which build core models, infrastructure, or data layers, account for just 2.8% of AI-native startups. The overwhelming majority instead build products on top of third-party foundation models. Applied vertical AI firms, which embed generative capabilities into products for specific industries or workflows, are the largest category at 59% of AI-native startups, followed by thick LLM integrators (13%), autonomous-agent firms (12.7%), and tooling firms (5.7%). Generative content platforms (3.4%) and thin LLM wrappers (3.2%) are comparatively rare. We also measure the extent to which AI-native startups depend on third-party AI APIs for their core product functionality (panel (b)). We classify 10.6% of AI-native startups as having high structural reliance on external AI models or APIs (high RAD score), and a further 80.8% as having medium reliance on external AI models (medium RAD), meaning that they use external AI services combined with proprietary workflow, data, or product layers. Only 8.6% have low dependence on third-party AI APIs (low RAD). In panel (c), reliance on third-party APIs is unsurprisingly higher for thin LLM wrappers (1C), which are essentially startups that provide sophisticated prompts. In contrast, startups that supply foundation layers and developing applied vertical AI have the lowest dependence on third-party APIs (1A and 1E, respectively). Dependence on third-party APIs further increases during the sample period (panel (d)). Thus, although a large share of AI-native entrants add nontrivial proprietary layers on top of external infrastructure, almost none operate independently of it.

Overall, these patterns indicate that GenAI-era entrepreneurship is built predominantly on foundation models supplied by a small number of incumbents rather than on models developed in-house, so that the core capabilities of most AI-native entrants depend on continued access to third-party APIs.

Startup viability. While it is notoriously hard to evaluate startups' economic viability for lack of financial and performance data, we consider an offline website to be a strong indication that the startup has exited.[11] This allows us to gauge the viability of the new firms whose entry has been spurred by GenAI. In Figure A.7, the failure rate, as gauged by an offline website in 2026, is mechanically decreasing with the startup cohort. More interestingly, within each cohort, a higher proportion of non-AI-enabled startups exits, suggesting that AI adoption is not associated with lower-quality startups, as implied by our conceptual framework. We reach similar conclusions in Figure A.8 for AI-native firms. Unsurprisingly, in this case, the failure rate is highest among thin LLM wrappers, that is, AI-native firms that only superficially adopt GenAI in their products (Figure A.9).

Taken together, it does not appear that GenAI is favoring the entry of low-quality ventures, less viable than ventures with more traditional businesses. Thus, having established that our proxy for industry exposure to GenAI is indeed related to entry and captures the new technology's adoption, we can explore how GenAI is changing entrepreneurial activity.

6 Geographic Dispersion of New Entrants

Human capital externalities and financial frictions concentrate entrepreneurial activity in locations that offer thick labor markets and investor networks (Ellison and Glaeser, 1997, Ewens

[11] Website activity is commonly used to measure startup survival in the entrepreneurial finance literature (e.g., Ewens and Townsend, 2020, Huang, Ivković, Jiang, and Wang, 2023, Hu and Ma, 2025). Maintaining an active website is a basic operational requirement for most startups, so deactivation is likely to accompany exit. Rebranding or domain changes may also make the original website inactive, although such changes may effectively represent the emergence of a new startup identity.

and Farre-Mensa, 2022). According to the first prediction of our conceptual framework, GenAI should erode agglomeration economies: GenAI provides on-demand expertise for tasks that can be augmented and automated. For this reason, founders may no longer need to be physically close to specialized human capital. Furthermore, since GenAI reduces the need to scale the labor force, new ventures' financing needs decrease. Entry should therefore rise more in locations that previously lacked access to thick labor markets and financing networks. Evidence that entrants concentrate further would instead point to a complementarity between GenAI and agglomeration, if access to frontier talent, compute, or proprietary data has become more valuable.

InfoGroup data provides firms' locations and allows us to explore how GenAI is affecting the geographic distribution of new ventures. Table 4 asks whether the number of geographic areas with entry in a given industry during a year increases post-GenAI and estimates the difference-in-differences model used in the previous tests to evaluate any changes. In addition to the number of geographic areas, we measure the concentration of new firms in a given industry and year using a geographic Herfindahl–Hirschman Index (HHI), defined as the sum of the squares of the shares of new entrants across geographic locations for that industry and year.

We find that firm entry becomes more geographically dispersed following the diffusion of GenAI. Across specifications, we increase the granularity of geographic areas, progressing from states (columns 1 and 4) to combined statistical areas (CSAs) in columns 2 and 5, to counties (columns 3 and 6). More states, CSAs, and counties record new firm formation in highly exposed industries after 2022 (columns 1 to 3), and the geographic HHI declines correspondingly (columns 4 to 6). The effects are not only statistically significant but also economically meaningful. For instance, the coefficient estimate in column 6 implies an over 16% more pronounced decrease in the county level concentration of entry in industries more exposed to GenAI.

Table 5 provides further support for the mechanisms driving the increasing geographic dispersion of entrepreneurial activity by testing whether the increase in entry is indeed driven by geographical areas in which startups benefit less from the presence of specialized human capital and the availability of funding. To do so, we measure geographic industry specialization by assessing the pre-GenAI

distribution of local employment. For each county and four-digit NAICS industry, we compute the share of workers employed in that industry relative to total county employment, using the 2020-2022 pre-treatment average. A high employment share indicates that the local labor market is disproportionately concentrated in the entrant's industry and therefore offers a relatively deeper pool of workers with industry-relevant skills and experience. We classify counties in the top decile of this distribution as having higher availability of specialized labor in the respective industry.[12] We then compare post-GenAI entry responses across areas with high versus low specialization. In Panel A of Table 5, we find that the rise in firm formation in GenAI-exposed industries is concentrated in geographic areas with low local labor specialization in the entrant's industry (columns 1 and 3).

To examine the external financing channel, we exploit the spatial concentration of venture capital activity and distinguish between VC-hub or non-hub states based on pre-GenAI venture-finance intensity. California, Massachusetts, and New York, which have the highest levels of VC activity, are identified as VC hubs. We then compare post-GenAI entry dynamics between VC-rich and VC-scarce areas. The results in Panel B of Table 5 show that the increase in new firm formation in GenAI-exposed industries is disproportionately larger in states with lower VC presence, suggesting that GenAI reduces entrepreneurs' dependence on local venture capital ecosystems and enables firm creation in regions previously constrained by financing access. As an alternative, we define venture capital hubs at the county level by aggregating the total dollar value of investments made by U.S.-based VC investors in each county using Crunchbase data and designate the top ten counties as VC hubs. We again find stronger GenAI-induced entry in counties with limited VC activity (Panel B of Table B.9).

Overall, these patterns are consistent with the implication of our conceptual framework that, by automating and augmenting tasks, GenAI reduces startups' need for specialized labor and external funding and enables entrepreneurial entry outside traditional investor network hubs.

[12] Results are robust to alternative thresholds, including the top 5th percentile and top quartile of the specialization distribution. They are also robust to defining specialization at the CSA rather than the county level; see Panel A of Table B.9.

7 GenAI and Job Creation

7.1 The Size and Organization of New Entrants

Our conceptual framework also implies that new entrants are smaller and staff a smaller number of distinct occupational roles, as founders can successfully perform a wider number of tasks thanks to GenAI. We examine these two dimensions with different datasets, beginning with the size distribution in InfoGroup and turning to the richer merged Crunchbase–Revelio sample to study workforce composition.

Using InfoGroup data, we count the number of new firms in an industry and year below a size threshold to test whether newly founded firms have become smaller in the post-GenAI period.[13] We present the results in Table 6. We find that newly created firms in more exposed industries tend to be smaller. The increase in entry is concentrated among firms with initial employment below the pre-treatment industry median (columns 1 and 3), while the probability of entry for firms above the median industry size and the number of firms above the median size remain unchanged (columns 2 and 4). Using absolute size thresholds (e.g., 3/5/10 employees) produces similar results (Table B.10). It thus appears that access to GenAI tools enables entrepreneurs to substitute technology for labor in the early stages of firm formation, reducing the need for large founding teams.

Our conceptual framework also implies that GenAI-exposed startups reduce the mix of occupations they bring in at the time of founding, as some functions are automated or augmented by GenAI. To test this, we turn to the merged Crunchbase–Revelio sample, which provides employee-level occupational data for a subset of entrants. This allows us to test whether the decline in size reflects a proportional contraction of the founding team or a reallocation across occupations consistent with GenAI substituting for some tasks and complementing others.

To examine how startups' characteristics evolve in industries differentially exposed to GenAI,

[13] Because InfoGroup employee counts may be estimated and measured with error (Makridis and Ohlrogge, 2017), we focus on size thresholds here and rely more heavily on the richer Crunchbase–Revelio sample for workforce size and composition analyses.

we estimate the following empirical model:

$$Y_{ft} = \alpha + \beta \left(AEI_f^{\text{pre}} \times Post_t \right) + \gamma AEI_f^{\text{pre}} + \theta Post_t + \delta_{y(t)} + \sum_{c \in \mathscr{C}} \phi_c D_{fc} + \varepsilon_{ft}, \tag{8}$$

where the dependent variable, Y_{ft}, is a firm-level outcome, capturing characteristics of the firm's occupational composition, measured at the end of the founding year. Thus, the unit of observation is a startup firm f founded in year-month t, with t running from January 2020 to December 2024. We consider the following three outcomes: the total number of employees by the end of the founding year (*# employee*), the number of unique occupations among the firm's employees by the end of the founding year (*# occupation*), and the HHI of occupational concentration within the firm, capturing within-firm occupation diversity (*HHI occupation*).

The variable AEI_f^{pre} is the firm's ex-ante GenAI exposure, defined in Section 4, based on its Crunchbase category groups. Because Crunchbase reports the founding month, we exploit variation in the timing of entry within the year: the indicator $Post_t$ takes the value one for startups founded from August 2022 onward and zero otherwise, consistent with the business formation regressions. Thus, coefficient β captures how the composition of a startup's founding workforce changes in the post-period as a function of its exposure to GenAI. As in the earlier tests, we compare startups with exposure above the median with those with exposure below the median. We include founding-year fixed effects ($\delta_{y(t)}$), where $y(t)$ denotes the calendar year of t, and industry category indicators (D_{fc}) that take the value of one if firm f is associated with category group c. Because firms in Crunchbase can belong to multiple industry category groups, there is no mutually exclusive partition on which to cluster standard errors at the category-group level. We therefore sort startups into 50 equal-sized bins of GenAI exposure, approximately the number of unique category groups, and cluster standard errors at this level.

Table 7 confirms that, relative to less-exposed peers, startups with higher ex-ante GenAI exposure founded from August 2022 onward have *fewer employees*. More importantly, we find that these startups have *fewer distinct occupations* and *higher occupational concentration (HHI)* among

their employees at the end of the founding year. The coefficient estimates are largely invariant to whether we include founding-year fixed effects or add category controls. The estimates are not only statistically significant but also economically meaningful. The average startup in our sample has 4.6 employees. Thus, the estimate in column 1 implies a nearly 25% decrease in the number of employees for startups with GenAI exposure above the median following the diffusion of GenAI. In column 3, the concentration of occupations in startups with GenAI exposure above the median increases by nearly 10% following the diffusion of GenAI.

Figure 5 provides a visual characterization of the effects of GenAI exposure on startup size. We plot the frequency of startups by employee count during the pre- and post-periods, distinguishing between high-GenAI-exposure industries (panel (a)) and low-GenAI-exposure industries (panel (b)). The frequency of startups with only one employee increases significantly in the post-period in high-GenAI industry categories. This pattern is not observed in low-GenAI-exposure industries, which experience an increase in the number of startups with more than 10 employees. This evidence confirms that sample selection is unlikely to drive our findings, since small startups are more likely to be underreported in the merged Crunchbase-Revelio sample in recent years.

7.2 Occupational Demand at Startups

Our conceptual framework predicts that startups whose entry is driven by GenAI shed jobs that can be automated. At the same time, by reducing the labor and financing required to launch a firm, GenAI enables specialists with strong core skills to become entrepreneurs. These entrants may continue to hire, or increase their demand for, workers in occupations where GenAI complements human expertise and raises productivity. Thus, GenAI can reduce overall team size while shifting employment toward occupations that are augmented by the technology.

To evaluate this empirical implication, we begin by examining how the newly created jobs vary with their occupational exposure to GenAI across startups in industries with different levels of exposure to GenAI. Occupations are grouped into three categories based on their AEI values: *high*

exposure ($AEI > 0.1$), *medium* exposure ($0.01 \leq AEI \leq 0.1$), and *low* exposure ($AEI < 0.01$). For each startup, we count the number of employees in each group as of the end of the founding year and use these counts as dependent variables in the same difference-in-differences framework used in the earlier tests. Table 8 shows that employment in high-AEI occupations remains roughly stable or slightly increases, while employment in medium- and low-AEI occupations declines significantly, with the largest absolute reduction occurring in low-AEI roles. Since low-AEI roles are only marginally automated or augmented by the new technology, this finding suggests that the nature of startups is changing: even within the same industry, GenAI enables the entry of startups that involve tasks whose productivity is enhanced by the new technology, thus increasing the demand for high-AEI roles.

This interpretation implies that startups whose entry is favored by GenAI should have a higher proportion of employees in occupations with tasks that can be augmented by the new technology, while decreasing their hiring for occupations whose tasks can be automated. The AEI dataset allows us to distinguish these occupations. Claude.ai conversations are classified as involving "Validation", "Task Iteration", "Learning", "Directive", and "Feedback Loop". Following Demirer, Horton, Immorlica, Lucier, and Shahidi (2026), we classify the first three categories as AI-augmented activities and the remaining as automated. As in the earlier tests, we aggregate O*NET tasks at the occupational level and compute the automation share of tasks in a given occupation as automation-oriented usage divided by the sum of automation- and augmentation-oriented usage; occupations above the sample median are classified as automation-oriented and those at or below it as augmentation-oriented. Consistent with our stylized model, Table 9 shows that among startups more exposed to GenAI, employment in occupations performing tasks that can be automated declines, while the share of employees performing tasks that can be augmented by GenAI increases.

Similar patterns emerge among startups that we directly identify as GenAI adopters. Figure A.10 shows that AI-enabled (AI-native) startups experience both a larger decline in employment and a larger increase in the share of occupations with high and medium GenAI exposure than non-AI-enabled (non-AI-native) startups. The changes in occupational composition are particularly

noticeable in AI-native startups, where the need for technical skills to develop new products is presumably more pronounced.

Figure 6 provides related descriptive evidence on the occupations that experienced the largest employment losses (panel (a)) and gains (panel (b)) at startups in the post-GenAI period relative to the earlier period. Software developers recorded the largest occupational gains, rising from a 6.6% employment share pre-GenAI to 11.3% following the diffusion of the new technology. Engineers and information technology project managers, which are vital for the adoption of the new technology, also experience occupational gains in startups. The largest decreases in employment shares are among chief executives, marketing managers, and general operations managers, roles whose relative importance may have been reduced by GenAI.

Importantly, demand for occupations with high GenAI exposure in startups appears fundamentally different from that in large, mature companies. While high-AEI occupations whose tasks can be augmented by GenAI appear complementary to the new technology in startups, established companies have been found to cut jobs in these occupations (Acemoglu, Autor, Hazell, and Restrepo, 2022, Brynjolfsson, Mitchell, and Rock, 2018, Hampole, Papanikolaou, Schmidt, and Seegmiller, 2025). Startups could thus represent a countervailing channel, mitigating concerns about widespread job displacement.

7.3 Aggregate Employment Effects in New Entrants

Our findings indicate that GenAI has increased firm entry while reducing startups' employee headcount. From an aggregate perspective, it is important to understand whether new firm formation mitigates concerns about widespread labor displacement and depressed labor demand, or instead accentuates them due to startups' smaller size.

We examine how the net impact of these two forces shapes job creation in startups relatively more exposed to GenAI using QWI, a nationally representative administrative measure of employment and earnings for very young firms (age 0 and 1) and other companies at the four-digit NAICS

quarter level. Applying the same difference-in-differences framework as in the previous sections, we compare employment and wages in more versus less-exposed industries before and after the diffusion of GenAI. Because QWI data are quarterly, the post-period indicator takes the value one from 2022Q4, the quarter of ChatGPT's release, onward. Table 10 shows that employment among the new firms rises by 7.4% in more exposed industries (column 1), while employment among older firms shows no significant change (column 2). The lack of change among established firms is consistent with existing literature and likely reflects incumbents' slow adoption of technology. We also test whether monthly earnings in industries more exposed to GenAI have increased, as this would suggest increased labor demand in newly started firms in these industries. Consistent with the conjecture that demand for labor of startups in industries exposed to GenAI increases, we find that average monthly earnings increase by about 5% for workers in very young firms in high-exposure industries in comparison to the average industry (column 3), whereas monthly earnings for workers in high-GenAI-exposure incumbent firms remain flat (column 4).

The findings that aggregate employment and monthly employee earnings at newly established firms increase, notwithstanding the smaller size of the startups, reveal a countervailing channel not emphasized in prior work: the surge in new entrants generates net employment expansion and higher wages among the youngest firms in industries most affected by GenAI, even though these firms are started with fewer employees than in the pre-GenAI period.

8 GenAI and the Characteristics of Founders

The last set of empirical implications of our conceptual framework regards the founders. Entrepreneurs have to perform many tasks and have been typically considered jack-of-all-trades, that is, individuals with balanced skills (Lazear, 2004). Since GenAI can substitute for or augment some of their skills, it may favor the entry of individuals who are specialized in the core business of the startup. We thus explore whether the founders of the startups that enter following the diffusion of GenAI are more likely to be specialized in the core tasks of their business.

We estimate a regression similar to equation (8), using startup-founder-level observations constructed from snapshots of startups founded between 2020 and 2024. We proxy for a founder's core skills using several measures of her industry experience at the six-digit NAICS level.[14] Table 11 shows that following the diffusion of the new technology, founders of startups in high-GenAI exposure industries are more likely to have at least three years of experience in the industry of the startup, to have ever worked in that industry, and to have spent a larger proportion of their working life in that industry. The magnitude is sizable: the probability that a founder has at least three years of experience in the startup's industry increases by about 6.7 percentage points, roughly 32% of the sample mean.

To evaluate the founders' domain knowledge in the startups' business, we also measure their past occupations' GenAI exposure. Table B.12 shows that founders of startups in more GenAI-exposed categories have higher exposure to GenAI in their past occupations following the diffusion of the new technology. Therefore, this pattern does not merely reflect the founder's tendency to start businesses in industries where they already have experience.

Whereas the analysis above focuses on founders' occupational history, we move beyond structured measures of founders' prior work experience to examine how founders describe themselves. Using an LLM-assisted pipeline described in Appendix D, we extract granular traits from founders' LinkedIn profiles around the founding year, organize them into a bottom-up taxonomy, and compare cluster shares before and after GenAI diffusion. We include traits that can be inferred from the founders' narratives and actions. Throughout the analysis, we control for the total number of traits extracted from each founder's profile.

Table 12 shows that the distribution of founder traits shifts toward technical, task-specific capabilities in AI and machine learning development and away from capital raising and product management, consistent with our conceptual framework that entry spurred by GenAI should be driven by founders with high core task capabilities. Among founders of startups in industries with above-median GenAI exposure, the count of traits in the AI/ML and GenAI cluster rises

[14] In Table B.11, estimates are qualitatively and quantitatively invariant at the four-digit NAICS level.

by 0.246 after 2022 (column 2), against a sample mean of 0.229, indicating that the share of founders describing themselves in terms of AI and machine learning capabilities more than doubles in more-exposed industries following the diffusion of the new technology. By contrast, traits associated with capital raising and deal mechanics, product development, managerial and sales roles, and operations and logistics decline differentially in more-exposed industries, with the largest declines for capital and investing (-0.183, column 1) and product management or development. The shifts in founders' self-described capabilities are consistent with the increase in entry we observe in regions outside traditional venture capital hubs, documented in Section 6. Founders de-emphasizing the capital-raising and deal-making skills that have historically been central to venture-backed entrepreneurship are plausibly less reliant on proximity to venture capital networks. These shifts are robust to the inclusion of industry-category indicators and founding-year fixed effects (columns 6–10), even though the decline in capital-raising traits is imprecisely estimated. Overall, the evidence points to a recomposition of the founder pool rather than secular trends in how founders in specific industries describe themselves on professional platforms. The patterns are also robust to excluding traits logically inferred from founders' narratives and actions from the trait count, as reported in Table B.13.

One might worry that GenAI changed how founders describe themselves rather than who becomes a founder. Even so, the founders' reported traits would remain informative about the characteristics of successful entrepreneurs. Furthermore, we observe similar patterns for founders' characteristics that are not self-descriptions and are not subject to the same concern, including the GenAI exposure of prior occupations and industry experience at the six-digit NAICS level.

A possible alternative explanation for founders' technical backgrounds is that individuals in GenAI-exposed occupations are among the most affected by technological disruption in established companies, which could have led them to pursue entrepreneurship. However, evidence from Table 13 is inconsistent with founders in high-GenAI-exposure startups being pushed into entrepreneurship by job displacement. Following the diffusion of the new technology, founders of high-GenAI-exposed startups are more likely to have been employed, before their transition into entrepreneurship,

by firms that were rapidly expanding employment (column 1). In contrast, column 2 shows no corresponding increase in the likelihood that founders come from downsizing employers, defined as firms in the bottom decile of the employment-growth distribution. Such a conclusion is further reinforced by the fact that the founders do not become more likely to suffer a pay cut when they start a new firm (column 4). [15]

We also consider the founder's compensation at their previous employer to assess the hypothesis that GenAI may be attracting lower-quality or less experienced individuals into entrepreneurship. We do not find any changes in prior-employment compensation for founders of businesses highly exposed to GenAI following the diffusion of the new technology (column 3). This suggests that GenAI is unlikely to be drawing lower-quality individuals into entrepreneurship.

Overall, the evidence is more consistent with opportunity-driven entry: founders are drawn into entrepreneurial activity by investment opportunities and lower entry costs, rather than pushed into it by displacement or weaker outside options. As is consistent with our conceptual framework, in industries where GenAI is most relevant, the individuals who become entrepreneurs combine technical AI capabilities with domain knowledge from past occupations to develop new ventures, without a deterioration in the quality of the marginal entrant.

9 Conclusion

We document that GenAI is spurring entrepreneurial activity, driven both by AI-native ventures building new products and AI-enabled firms that lower their entry costs by automating and augmenting internal operations. By automating support activities and augmenting core ones, GenAI allows a founder to reach a quality that previously required either balanced skills or the funds to hire team members complementing the founder. Entry thus becomes more geographically dispersed, and the increase is concentrated in areas with thin specialized labor markets and limited venture

[15] Revelio Labs estimates salaries from job and employer characteristics using visa filings, self-reported salaries, and job postings. These estimates may differ from realized earnings, so we interpret the evidence as suggestive.

capital, indicating that GenAI substitutes for two of the agglomeration forces that have historically tied entrepreneurship to a few locations. Consistent with this mechanism, new founders are more specialized in technical skills related to their venture and less defined by the balanced, generalist profile that has traditionally characterized entrepreneurs. We find no evidence that these founders are of lower quality, as their compensation before entry does not decline.

Although individual entrants are smaller, they are no less viable than earlier cohorts, and the rise in entry more than offsets their smaller scale: aggregate employment and earnings increase among young firms in exposed industries. Furthermore, startups appear to have higher demand for individuals with prior experience in occupations with high exposure to GenAI, which are being shed at mature companies. New ventures, therefore, emerge as a margin through which GenAI may expand labor demand. These results alleviate concerns about widespread job displacement but also underscore the importance of technical skills and prior experience. Specifically, high-level AI education (Gofman and Jin, 2024) and training programs for new graduates may be crucial for the economy to take full advantage of the countervailing channel we highlight.

Importantly, we also document how new entrants adopt the new technology. Almost all AI-native entrants build on externally supplied foundation models rather than on models developed in-house, so their core capabilities depend on continued access to third-party APIs. Policymakers aiming to leverage the new technology to spur growth and employment should focus on ensuring reliable access to foundational models. Because GenAI substitutes for local investor networks, policies aimed at broadening regional participation in entrepreneurship may also depend less on building local venture capital ecosystems than has been assumed. Reducing compliance burdens and legal uncertainties for small businesses relying on GenAI, for instance, through regulatory sandboxes, may further enable economies to reap the benefits of the new technology.

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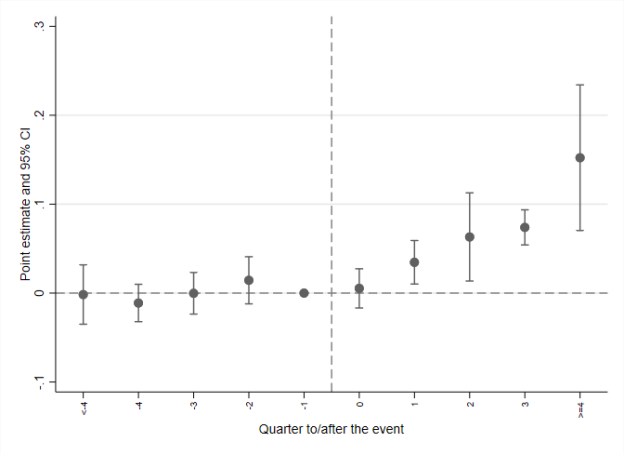
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Figure 1
GenAI Introduction and Corporate Business Applications



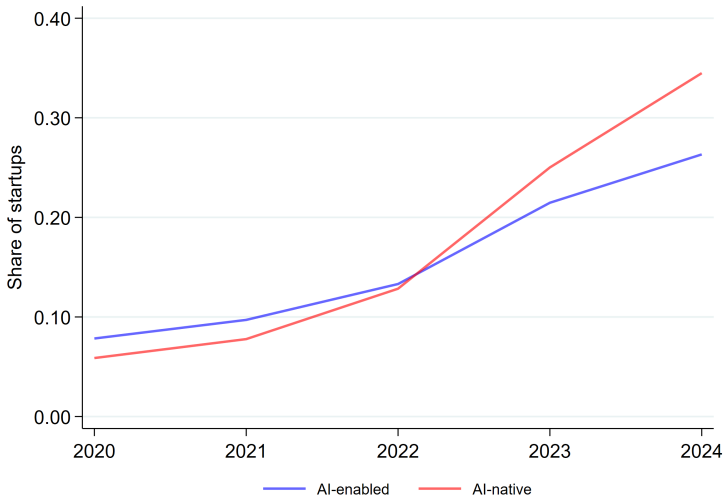
(a) Raw patterns by exposure



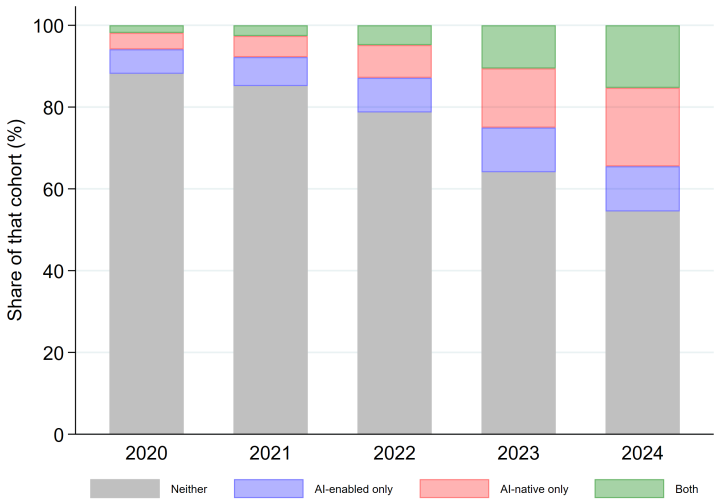
(b) Regression dynamics

This figure examines the effect of GenAI introduction on corporate business applications (CBA). Panel (a) plots the raw monthly series, separately for sectors with above- and below-median GenAI exposure. Each series is the group mean of the log number of applications, normalized to its own value in July 2022 and multiplied by 100, the month before the release of Stable Diffusion; the two vertical lines mark the releases of Stable Diffusion (August 2022) and ChatGPT (November 2022). The sample begins in January 2020. Panel (b) plots event-time coefficients from the dynamic specification corresponding to Column (3) in Table 2. To reduce estimation noise, months are aggregated into quarterly intervals. Quarter -1 corresponds to the second quarter of 2022 and serves as the omitted baseline period. Standard errors are Driscoll–Kraay with a bandwidth of four monthly lags.

Figure 2
The Diffusion of AI-Enabled and AI-Native Startups



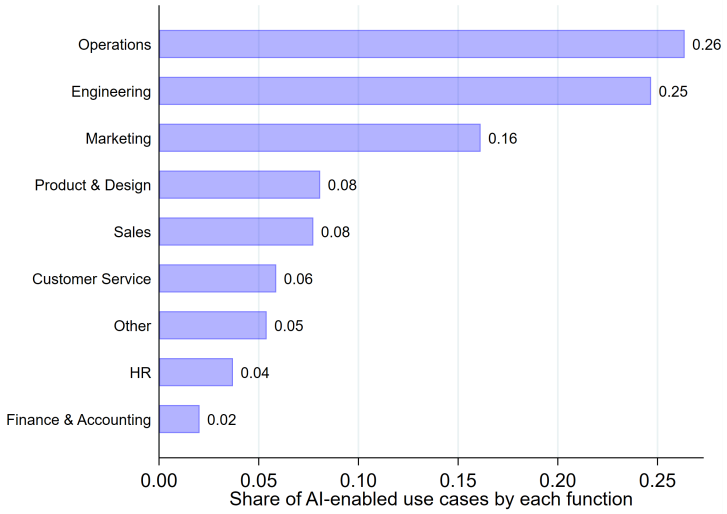
(a) Share of startups by classification



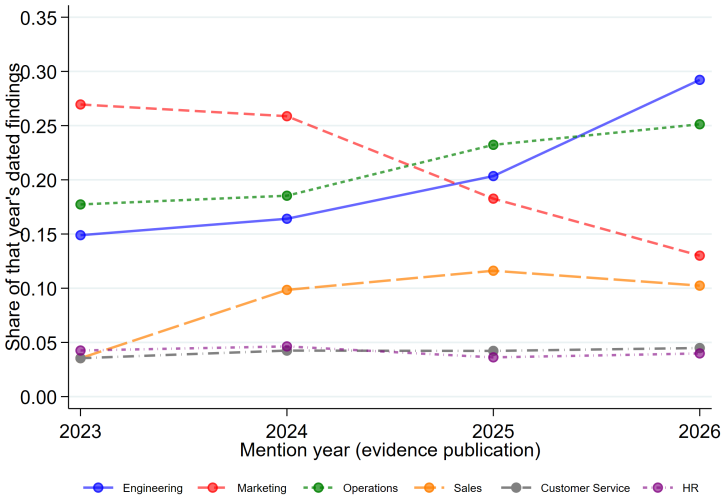
(b) Joint composition across the two classifications

This figure plots the diffusion of the two AI classifications across founding cohorts. A startup is *AI-enabled* if the deep-research pipeline returns at least one verified finding of internal GenAI use, and *AI-native* if its website text identifies AI as the product itself. Both panels use all 44,387 startups in the Crunchbase–Revelio sample as the denominator, so that firms without classifiable evidence count as neither AI-enabled nor AI-native. Panel (a) plots the share of each founding cohort classified as AI-enabled and as AI-native. Panel (b) splits each cohort into four mutually exclusive groups defined by the two classifications jointly: neither, AI-enabled only, AI-native only, and both. Bars sum to one hundred percent within each cohort.

Figure 3
Business Functions Underlying AI-Enabled Startups



(a) Business functions

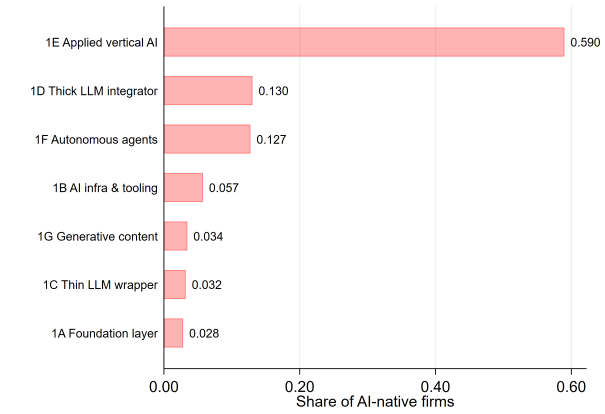


(b) Function shares by mention year

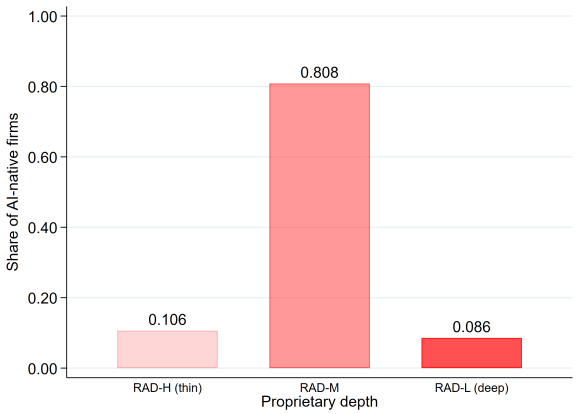
This figure summarizes the business functions in which AI-enabled startups use GenAI. Panel (a) reports the share of AI-enabled use cases attributed to each business function, pooling all verified findings. Panel (b) plots the share of each year's dated findings attributed to each business function over mention years 2023–2026, where the mention year is the publication date of the underlying evidence. Shares in Panel (a) sum to one across the categories shown; shares in Panel (b) sum to one within each mention year.

Figure 4

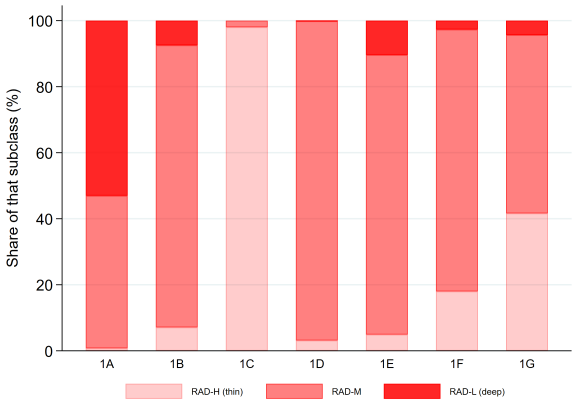
The Composition of AI-Native Startups: Product Subclass and Proprietary Depth



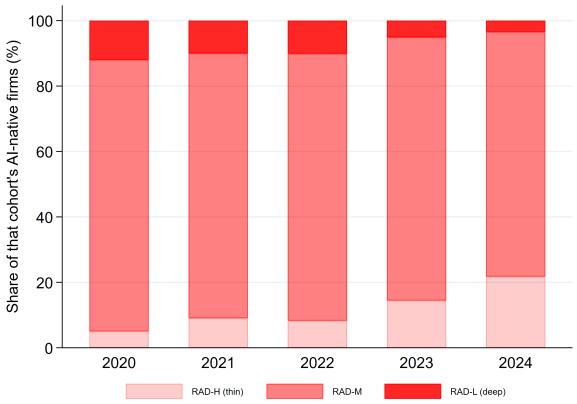
(a) Product subclass



(b) Proprietary depth



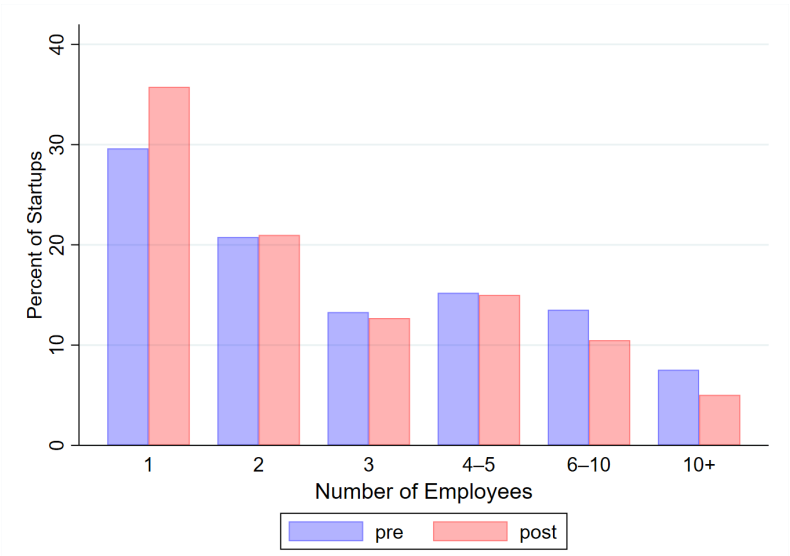
(c) Proprietary depth within subclass



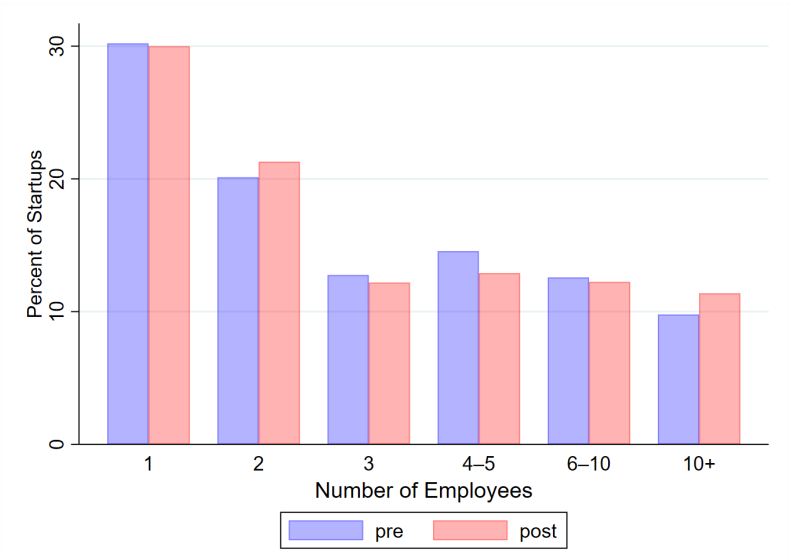
(d) Proprietary depth by founding cohort

This figure characterizes the AI-native startups, along two dimensions assigned by the website-text classifier described in Appendix C: the product subclass the firm occupies, and its structural dependence on third-party model APIs (the RAD score). Subclasses are 1A foundation layer, 1B AI infrastructure and tooling, 1C thin LLM wrapper, 1D thick LLM integrator, 1E applied vertical AI, 1F autonomous agents, and 1G generative content. The RAD score takes three values: RAD-H denotes high structural dependence on external models, with little proprietary layer of the firm's own (thin); RAD-M denotes partial independence, combining third-party models with some proprietary component; and RAD-L denotes low dependence, with substantial in-house model development (deep). Panel (a) plots the share of AI-native firms in each subclass. Panel (b) plots the share of AI-native firms at each level of proprietary depth. Panel (c) plots the proprietary depth composition within each subclass, with bars summing to one hundred percent by subclass. Panel (d) plots the proprietary depth composition of each founding cohort from 2020 to 2024, with bars summing to one hundred percent by cohort. Firms are classified from live website text where the site is reachable at the May 2026 crawl and from the last archived snapshot otherwise, so composition is comparable across cohorts and is not conditioned on survival.

Figure 5
GenAI Introduction and the Changing Team Size Distribution of Startups



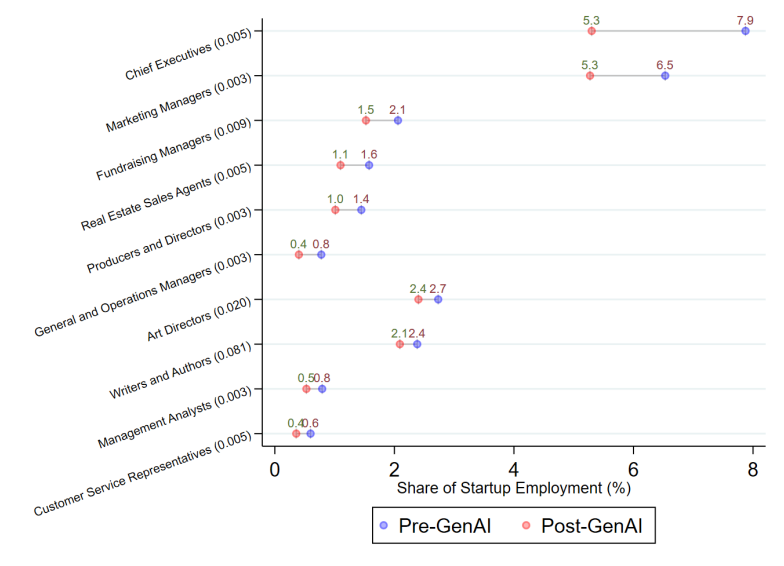
(a) High GenAI Exposure



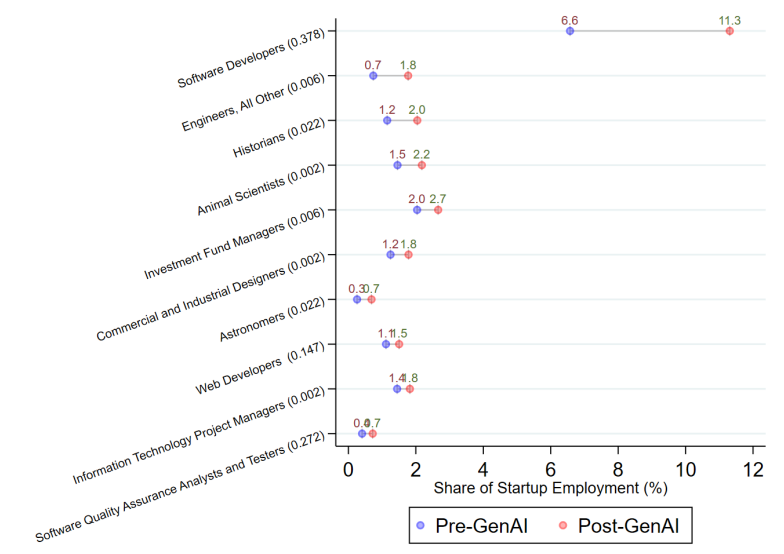
(b) Low GenAI Exposure

This figure shows the distribution of startups by team size—measured as the number of employees at the end of the founding year—before and after the introduction of GenAI, using the Crunchbase–LinkedIn merged sample. Panel (a) presents startups in industries with high GenAI exposure (above median AEI), while Panel (b) presents those in low-exposure industries.

Figure 6
GenAI Introduction and Changing Startup Occupations



(a) Losing Occupations



(b) Gaining Occupations

This figure shows the six-digit SOC occupations that experienced the largest changes in their employment shares among startups following the introduction of GenAI, using the Crunchbase–LinkedIn merged sample. Panel (a) lists the ten occupations with the largest declines in employment share, while Panel (b) lists those with the largest gains.

Table 1
Summary Statistics

Panel A: Sector-Year-Month Level (BFS)

Variable	N	Mean	Std. Dev.	Median	p1	p25	p75	p99
# BA	1,105	23,961	20,311	21,052	362	8,162	34,412	85,976
# HBA	1,105	8,076	7,969	5,190	54	2,051	15,215	26,770
# CBA	1,105	2,838	2,704	2,335	30	1,030	3,837	8,575
# PBF4Q	1,099	1,528	1,269	1,159	15	476	2,668	4,142
# PBF8Q	1,099	2,109	1,746	1,547	21	669	3,749	5,658
AEI	1,105	0.009	0.011	0.005	0.001	0.003	0.008	0.039

Panel B: Industry-Year Level (aggregated from InfoGroup)

Variable	N	Mean	Std. Dev.	Median	p1	p25	p75	p99
I (firm entry)	1,430	0.778	0.416	1.000	0.000	1.000	1.000	1.000
# firm entry	1,430	63.096	169.319	7.000	0.000	1.000	45.000	741.000
# firm entry (below median size)	1,430	34.311	98.802	3.000	0.000	0.000	23.000	453.000
# firm entry (above median size)	1,430	28.785	82.546	2.000	0.000	0.000	18.000	368.000
# state	1,430	12.779	14.972	5.000	0.000	1.000	22.000	49.000
# csa	1,430	17.481	25.544	5.000	0.000	1.000	24.000	103.000
# county	1,430	41.479	84.113	6.000	0.000	1.000	41.000	394.000
HHI state	1,430	0.441	0.408	0.222	0.042	0.080	1.000	1.000
HHI csa	1,430	0.458	0.403	0.250	0.053	0.093	1.000	1.000
HHI county	1,430	0.408	0.429	0.167	0.006	0.028	1.000	1.000
AEI	1,430	0.009	0.015	0.004	0.001	0.002	0.008	0.073

Panel C: Startup Level (Crunchbase-LinkedIn Merged Sample)

Variable	N	Mean	Std. Dev.	Median	p1	p25	p75	p99
# employee	42,537	4.592	7.325	2.000	1.000	1.000	5.000	39.000
# occupation	42,537	3.345	3.839	2.000	1.000	1.000	4.000	20.000
HHI occupation	42,537	0.570	0.333	0.500	0.074	0.280	1.000	1.000
# employee (high AEI)	42,537	0.434	1.333	0.000	0.000	0.000	0.000	5.000
# employee (mid AEI)	42,537	0.769	1.648	0.000	0.000	0.000	1.000	7.000
# employee (low AEI)	42,537	3.389	6.045	2.000	0.000	1.000	4.000	31.000
Startup AEI	42,537	0.044	0.016	0.039	0.019	0.031	0.058	0.074

Panel D: Founder Level (Crunchbase-LinkedIn Merged Sample)

Variable	N	Mean	Std. Dev.	Median	p1	p25	p75	p99
Prior AEI (most recent)	15,236	0.046	0.111	0.005	0.000	0.003	0.019	0.378
Prior AEI (mean)	15,236	0.047	0.094	0.008	0.000	0.004	0.023	0.378
Prior AEI (max)	15,236	0.092	0.157	0.020	0.000	0.006	0.062	0.588
$\geq$3 Years in Startup Industry (NAICS 6)	15,695	0.212	0.409	0.000	0.000	0.000	0.000	1.000
Ever in Startup Industry (NAICS 6)	15,695	0.394	0.489	0.000	0.000	0.000	1.000	1.000
Share of Years in Startup Industry (NAICS 6)	14,843	0.167	0.280	0.000	0.000	0.000	0.226	1.000
# Traits: Capital/Investing	15,495	0.477	1.043	0.000	0.000	0.000	1.000	5.000
# Traits: AI/ML & GenAI	15,495	0.228	0.660	0.000	0.000	0.000	0.000	3.000
# Traits: Product Mgmt/Development	15,495	0.463	0.840	0.000	0.000	0.000	1.000	4.000
# Traits: Marketing/Sales	15,495	0.785	1.427	0.000	0.000	0.000	1.000	6.000
# Traits: Operations/Logistics	15,495	1.326	1.726	1.000	0.000	0.000	2.000	7.000

This table reports summary statistics for the four samples used in the empirical analysis. Panel A summarizes business registration counts from the Census Bureau's Business Formation Statistics (BFS), at the sector (two-digit NAICS)–year–month level. Panel B summarizes industry–year–level variables constructed from the InfoGroup firm panel, aggregated to the four-digit NAICS level. Panel C summarizes startup-level variables from the Crunchbase–LinkedIn merged sample, describing the size and occupational composition of startups' workforces at founding. Panel D summarizes founder-level characteristics from the same Crunchbase–LinkedIn sample. AEI denotes the AI exposure index, measured at the level of the observation in each panel. Sample sizes vary across variables within D because of missing values in the underlying LinkedIn profiles.

Table 2
GenAI Introduction and Business Formation

	(1)	(2)	(3)	(4)	(5)
Dep. Var.	# new business formation (log)				
	BA	HBA	CBA	PBF4Q	PBF8Q
AEI (above median) × Post	0.020	0.083***	0.120***	0.075***	0.081**
	[0.021]	[0.023]	[0.033]	[0.028]	[0.031]
Sector FE	YES	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES	YES
Obs	1,105	1,105	1,105	1,099	1,099
Adj R2	0.993	0.993	0.988	0.988	0.984

This table reports the estimated effects of GenAI introduction on new business formation using aggregate-level data from the U.S. Census Bureau's *Business Formation Statistics* (BFS). The outcome variable is the logarithm of the number of new business formations in a two-digit NAICS industry, measured monthly. The sample covers January 2020 to May 2025. AEI (above median) is a dummy that equals one for sectors whose AI exposure, measured by the Anthropic Economic Index (AEI) aggregated to the two-digit NAICS level, is above median, and zero otherwise. *Post* is a dummy that equals one from August 2022 onward and zero otherwise. Columns (1)-(5) use five BFS outcome measures: (1) all Business Applications (BA), (2) High-Propensity Business Applications (HBA), (3) Business Applications from Corporations (CBA), (4) Projected Business Formations within four quarters (PBF4Q), and (5) Projected Business Formations within eight quarters (PBF8Q). HBA represent EIN applications with characteristics historically associated with the successful establishment of active firms, such as intended payroll or corporate registration. PBF measures are Census-derived projections of business applications expected to become active employer firms within four to eight quarters and are considered a leading indicator of realized business creation. Sector (two-digit NAICS) and month-year fixed effects are included in all specifications. Standard errors are Driscoll–Kraay, which are robust to correlation across sectors within a month and to serial correlation within a sector, with a bandwidth of four monthly lags. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 3
GenAI Introduction and Firm Entry

	(1)	(2)	(3)	(4)
Dep. Var.	I (firm entry)		# firm entry	
AEI (quartile) × Post	0.040**		0.055*	
	[0.017]		[0.033]	
AEI (above median) × Post		0.111***		0.212**
		[0.037]		[0.089]
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430
Adj R2	0.471	0.473	0.888	0.888

This table reports the estimated effects of GenAI introduction on new firm entry using establishment-level data aggregated from the *InfoGroup* firm panel. The unit of observation is a four-digit NAICS industry–year cell. The sample includes newly established, single-location firms founded between 2020 and 2024. The dependent variable in Columns (1) and (2) is an indicator for any firm entry, or *I(firm entry)*. The dependent variable in Columns (3) and (4) is the logarithm of one plus the number of firm entries in each industry–year cell, or *# firm entry*. *Post* is a dummy that equals one for 2023–2024 and zero otherwise. *AEI (quartile)* denotes the four-digit NAICS-level AI exposure measure based on AEI, divided into quartiles according to the exposure distribution across industries. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year fixed effects are included in all specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 4
GenAI Introduction and the Geographic Distribution of Firm Entry

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	# state	# csa	# county	HHI state	HHI csa	HHI county
AEI (above median) × Post	0.204***	0.197***	0.221***	-0.064**	-0.058**	-0.068***
	[0.069]	[0.072]	[0.084]	[0.025]	[0.024]	[0.026]
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430	1,430	1,430
Adj R2	0.854	0.868	0.884	0.764	0.762	0.776

This table reports the estimated effects of GenAI introduction on the geographic distribution of new firm entries, using establishment-level data aggregated from the InfoGroup firm panel. The unit of observation is a four-digit NAICS industry–year cell. The sample includes newly established, single-location firms founded between 2020 and 2024. Columns (1)-(3) examine the extensive margin of geographic dispersion. The dependent variables are measured as the logarithm of one plus the number of distinct geographic units—states, combined statistical areas (CSAs), and counties—recording new firm entries within each industry–year cell. Columns (4)-(6) examine geographic concentration, measured by the Herfindahl–Hirschman Index (HHI) of firm entry across states, CSAs, and counties. *Post* is a dummy that equals one for 2023–2024 and zero otherwise. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year fixed effects are included in all specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 5
Channels Behind Dispersed Entry: Labor Specialization and Financing

Panel A: Relaxed Labor Specialization Requirements

	(1)	(2)	(3)	(4)
Dep. Var.	I (firm entry)		# firm entry (log)	
	non-specialized	specialized	non-specialized	specialized
AEI (above median) × Post	0.110***	-0.028	0.226**	-0.155***
	[0.037]	[0.037]	[0.089]	[0.059]
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430
Adj R2	0.48	0.571	0.886	0.85

Panel B: Relaxed Financing Requirements

	(1)	(2)	(3)	(4)
Dep. Var.	I (firm entry)		# firm entry (log)	
	non VC hubs	VC hubs	non VC hubs	VC hubs
AEI (above median) × Post	0.101***	0.057	0.224***	-0.006
	[0.038]	[0.037]	[0.083]	[0.077]
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430
Adj R2	0.489	0.587	0.89	0.86

This table examines two channels through which the introduction of GenAI may have contributed to more geographically dispersed firm entry, using establishment-level data aggregated from the InfoGroup firm panel. The unit of observation is a four-digit NAICS industry–year cell. The sample includes newly established, single-location firms founded between 2020 and 2024. Panel A investigates the labor specialization channel. Each county is classified as having *specialized* or *non-specialized* labor based on the pre-GenAI distribution of local employment. For each industry, the share of workers employed in that industry relative to total county employment is calculated, and counties in the top decile of this distribution are defined as specialized, while all others are classified as non-specialized. Panel B tests the venture capital financing channel. We classify regions into *VC hubs* and *non-hubs* based on pre-GenAI venture investment intensity. VC hubs correspond to the three states with the highest levels of venture capital activity—California, Massachusetts, and New York—while all other states are treated as non-hubs. The dependent variable is an indicator for any firm entry, *I(firm entry)*, in Columns (1) and (2), and the logarithm of one plus the number of new firm entries, *# firm entry*, in Columns (3) and (4), each measured at the industry–year cell level and computed separately for entry in non-specialized (Columns (1) and (3)) or specialized (Columns (2) and (4)) counties in Panel A, and in non-hub (Columns (1) and (3)) or hub (Columns (2) and (4)) states in Panel B. *Post* is a dummy that equals one for 2023–2024 and zero otherwise. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year fixed effects are included in all specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 6
GenAI Introduction and the Size Composition of Firm Entry

	(1)	(2)	(3)	(4)
Dep. Var.	I (firm entry)		# firm entry	
	below median size	above median size	below median size	above median size
AEI (above median) × Post	0.108***	0.046	0.231***	-0.094
	[0.040]	[0.032]	[0.082]	[0.072]
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430
Adj R2	0.468	0.628	0.857	0.916

This table reports the estimated effects of GenAI introduction on the size distribution of new firms, using establishment-level data aggregated from the *InfoGroup* firm panel. The unit of observation is a four-digit NAICS industry–year cell. The sample includes newly established, single-location firms founded between 2020 and 2024. The dependent variable is an indicator for any firm entry, *I(firm entry)*, in Columns (1) and (2), and the logarithm of one plus the number of new firm entries, *# firm entry*, in Columns (3) and (4), each measured at the industry–year cell level and computed separately for entrants with initial employment below (Columns (1) and (3)) or above (Columns (2) and (4)) the pre-treatment (2020–2022) industry median. *Post* is a dummy that equals one for 2023–2024 and zero otherwise. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year fixed effects are included in all specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 7

GenAI Introduction and Startup Team Size and Occupational Breadth of Founding Teams

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	# employee	# occupation	HHI occupation	# employee	# occupation	HHI occupation
AEI (above median) × Post	-1.107***	-0.560***	0.051***	-0.978***	-0.485***	0.045***
	[0.256]	[0.133]	[0.011]	[0.266]	[0.139]	[0.012]
Industry Category Indicators	NO	NO	NO	YES	YES	YES
Year FE	NO	NO	NO	YES	YES	YES
Obs	42,537	42,537	42,537	42,537	42,537	42,537
Adj R2	0.003	0.003	0.002	0.011	0.018	0.018

This table presents estimated effects of GenAI introduction on startup team size and occupational composition, using firm-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is a startup firm founded between 2020 and 2024, with all dependent variables measured at the end of the founding year. The dependent variable is the total number of employees in Columns (1) and (4), the number of unique eight-digit O*NET-SOC occupations within the startup in Columns (2) and (5), and the Herfindahl–Hirschman Index (HHI) of employment concentration across occupations in Columns (3) and (6). *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. Columns (1)–(3) exclude startup category-group controls and year fixed effects, while Columns (4)–(6) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 8
GenAI Introduction and Startup Employment by Occupational Exposure

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	# employee (by occupation's AEI exposure)					
	high AEI	mid AEI	low AEI	high AEI	mid AEI	low AEI
AEI (above median) × Post	0.068	-0.285***	-0.890***	0.022	-0.241***	-0.760***
	[0.044]	[0.053]	[0.222]	[0.040]	[0.053]	[0.228]
Dep. Var. Mean	0.434	0.769	3.389	0.434	0.769	3.389
Industry Category Indicators	NO	NO	NO	YES	YES	YES
Year FE	NO	NO	NO	YES	YES	YES
Obs	42,537	42,537	42,537	42,537	42,537	42,537
Adj R2	0.047	0.001	0.012	0.064	0.037	0.025

This table reports the estimated effects of GenAI introduction on startup employment across occupations with differing exposure to GenAI-related tasks, using firm-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is a startup firm founded between 2020 and 2024, with all dependent variables measured at the end of the founding year. Occupations are grouped into three categories based on their Anthropic Economic Index (AEI) values: high exposure ($AEI > 0.1$) in Columns (1) and (4), medium exposure ($0.01 \leq AEI \leq 0.1$) in Columns (2) and (5), and low exposure ($AEI < 0.01$) in Columns (3) and (6). The dependent variable is the number of employees in each AEI exposure category within a firm. *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. Columns (1)–(3) exclude startup category-group controls and year fixed effects, while Columns (4)–(6) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 9
GenAI Introduction and Startup Employment: Automation vs. Augmentation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dep. Var.	# employee		emp. share		# employee		emp. share	
	automation	augmentation	automation	augmentation	automation	augmentation	automation	augmentation
AEI (above median) × Post	-0.179***	-0.038	0.006	0.046***	-0.153***	-0.066	0.010	0.032***
	[0.042]	[0.054]	[0.009]	[0.012]	[0.042]	[0.050]	[0.009]	[0.009]
Dep. Var. Mean	0.644	0.559	0.157	0.127	0.644	0.559	0.157	0.127
Industry Category Indicators	NO	NO	NO	NO	YES	YES	YES	YES
Year FE	NO	NO	NO	NO	YES	YES	YES	YES
Obs	42,537	42,537	42,537	42,537	42,537	42,537	42,537	42,537
Adj R2	0.000	0.034	0.002	0.082	0.046	0.055	0.060	0.128

This table reports the estimated effects of GenAI introduction on the automation vs. augmentation oriented composition of startup employment, using firm-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is a startup firm founded between 2020 and 2024, with all dependent variables measured at the end of the founding year. Occupation-level GenAI usage shares for different purposes are taken from the Anthropic Economic Index taxonomy of interaction patterns. Automation-oriented usage comprises the *directive* and *feedback loop* patterns, in which the user delegates a task and the model produces the output, either in a single exchange or by iterating against environmental signals rather than against the user. Augmentation-oriented usage comprises the *task iteration*, *validation*, and *learning* patterns, in which the human remains in the loop, co-producing drafts, checking work performed by the user, or building the user's own capability. Filtered interactions are excluded from both aggregates. For each occupation in the classification, we compute the automation share as automation-oriented usage divided by the sum of automation- and augmentation-oriented usage; occupations above the sample median are classified as automation-oriented and those at or below it as augmentation-oriented. The sample of occupations is restricted to those with meaningful GenAI exposure ($AEI \geq 0.01$), below which measured usage is too sparse for the automation/augmentation orientation split to be informative. The dependent variable is the number of employees in each orientation group in Columns (1)–(2) and (5)–(6), and that group's share of total firm employment in Columns (3)–(4) and (7)–(8). Columns (1)–(4) report results without startup category-group controls or year fixed effects, while Columns (5)–(8) include both. *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 10
GenAI Introduction and Aggregate Employment and Earnings by Firm Age

	(1)	(2)	(3)	(4)
Dep. Var.	employment		employee earnings	
Firm Age	0−1	>1	0−1	>1
AEI (above median) × Post	0.074*	0.034	222.390**	-20.9
	[0.043]	[0.023]	[106.796]	[51.887]
Dep. Var. Mean	8.153	9.352	4624.199	5248.697
Industry FE	YES	YES	YES	YES
Year-Quarter FE	YES	YES	YES	YES
Obs	5,218	20,890	5,211	20,880
Adj R2	0.973	0.553	0.829	0.801

This table reports the estimated effects of GenAI introduction on industry-level employment and earnings using the Census Quarterly Workforce Indicators (QWI). The unit of observation is a four-digit NAICS industry-year-quarter. The dependent variable is the logarithm of employment in Columns (1) and (2), and average monthly earnings in Columns (3) and (4), each computed separately for age 0–1 firms (Columns (1) and (3)) and all older firms in the industry (Columns (2) and (4)). *Post* is a dummy that equals one from 2022Q4 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year-quarter fixed effects are included in all specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 11
GenAI Introduction and Startup Founders' Experience in the Startup Industry

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	≥3 Years in Startup Industry	Ever in Startup Industry	Share of Years in Startup Industry	≥3 Years in Startup Industry	Ever in Startup Industry	Share of Years in Startup Industry
AEI (above median) × Post	0.067**	0.075**	0.033**	0.069**	0.077**	0.035**
	[0.025]	[0.031]	[0.016]	[0.026]	[0.030]	[0.016]
Dep. Var. Mean	0.212	0.394	0.167	0.212	0.394	0.167
Industry Category Indicators	NO	NO	NO	YES	YES	YES
Year FE	NO	NO	NO	YES	YES	YES
Obs	15,236	15,236	14,407	15,236	15,236	14,407
Adj R2	0.008	0.013	0.005	0.020	0.025	0.010

This table reports the estimated effects of GenAI introduction on founders' experience in the startup industry, defined at the NAICS 6-digit level, using founder-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is an individual founder of startups founded between 2020 and 2024. The dependent variables measure the founder's prior work experience in the startup's own 6-digit NAICS industry: an indicator for at least three years of prior experience (*≥3 Years in Startup Industry*); an indicator for any prior experience (*Ever in Startup Industry*); and the share of the founder's career spent in that industry (*Share of Years in Startup Industry*). *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. Columns (1)–(3) report estimates without startup category-group controls or year fixed effects, while Columns (4)–(6) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 12
GenAI Introduction and Startup Founders' Traits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dep. Var.	Capital/ Investing	AI/ML GenAI	Product Manager/ Development	Marketing/ Sales	Operations/ Logistics	Capital/ Investing	AI/ML GenAI	Product Manager/ Development	Marketing/ Sales	Operations/ Logistics
AEI (above median) × Post	-0.183**	0.246***	-0.078*	-0.103	-0.105	-0.100	0.079**	-0.102**	-0.062	-0.072
	[0.080]	[0.051]	[0.044]	[0.096]	[0.081]	[0.066]	[0.034]	[0.046]	[0.078]	[0.076]
Dep. Var. Mean	0.475	0.229	0.463	0.782	1.321	0.475	0.229	0.463	0.782	1.321
Industry Category Indicators	NO	NO	NO	NO	NO	YES	YES	YES	YES	YES
Year FE	NO	NO	NO	NO	NO	YES	YES	YES	YES	YES
Obs	15,154	15,154	15,154	15,154	15,154	15,154	15,154	15,154	15,154	15,154
Adj R2	0.052	0.080	0.149	0.162	0.324	0.157	0.177	0.167	0.252	0.351

This table reports the estimated effects of GenAI introduction on founders' trait counts, using founder-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is an individual founder of startups founded between 2020 and 2024. The dependent variables are counts of observed and inferred traits in five trait clusters defined in Appendix D: Cluster 3, capital, investing, and deal mechanics; Cluster 8, AI/ML and GenAI; Cluster 5, product building and "0–1" creation; Cluster 4, go-to-market, including marketing, sales, growth, and partnerships; and Cluster 12, operations, service delivery, and logistics. *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. All columns control for the total number of observed and inferred traits extracted for the founder across the 18 trait groups. Columns (1)–(5) report estimates without startup category-group controls or year fixed effects, while Columns (6)–(10) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 13
GenAI Introduction and Startup Founder Previous Employment and Compensation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dep. Var.	Expand	Shrink	Compensation	Paycut	Expand	Shrink	Compensation	Paycut
AEI (above median) × Post	0.046**	0.025	-0.012	-0.021	0.040**	0.027	0.019	-0.025
	[0.019]	[0.019]	[0.066]	[0.038]	[0.020]	[0.020]	[0.066]	[0.037]
Dep. Var. Mean	0.100	0.098	12.307	0.209	0.100	0.098	12.307	0.209
Industry Category Indicators	NO	NO	NO	NO	YES	YES	YES	YES
Year FE	NO	NO	NO	NO	YES	YES	YES	YES
Obs	8,826	8,826	14,284	12,343	8,826	8,826	14,284	12,343
Adj R2	0.001	0.004	0.002	0.010	0.004	0.009	0.013	0.012

This table reports the estimated effects of GenAI introduction on startup founders' previous-job employment and compensation characteristics using founder-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is an individual founder of startups founded between 2020 and 2024. *Expand* is an indicator for whether the founder's previous employer belongs to the top decile of the employment-growth distribution over the 12-month window surrounding the founder's job transition, while *Shrink* is an indicator for whether the previous employer belongs to the bottom decile. *Compensation* is the log of estimated total compensation in the founder's previous job, where compensation measures are obtained from Revelio Labs and are estimated using employee profiles, job titles, locations, and other worker characteristics. *Paycut* is an indicator equal to one if estimated total compensation at the startup is lower than estimated total compensation in the founder's previous job. *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. Columns (1)–(4) report estimates without startup category-group controls or year fixed effects, while Columns (5)–(8) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

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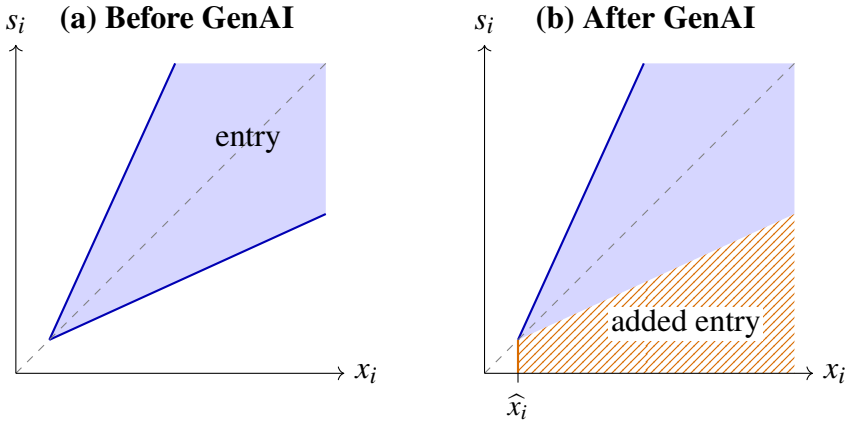
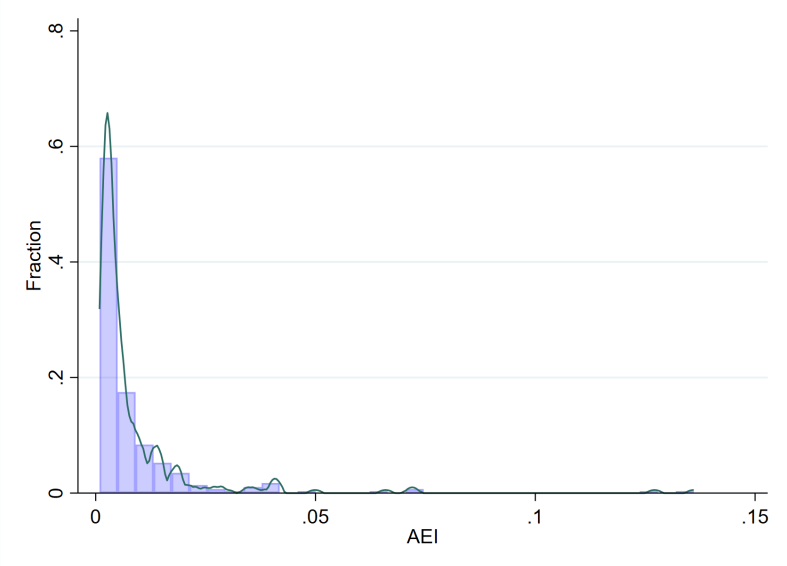
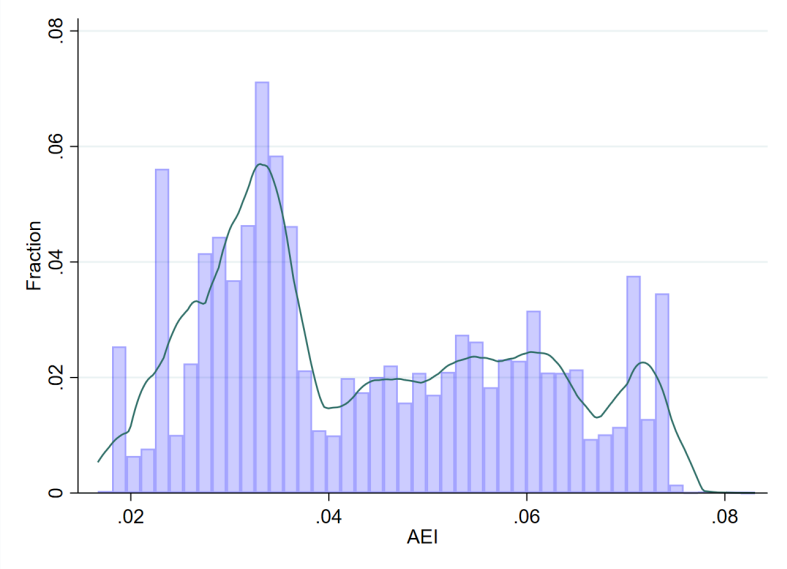


Figure A.1
Entrepreneurial entry in skill space. The figure depicts the entry decision of a solo, unfunded founder, as in Section 2.2. Before GenAI, entrepreneurs occupy a cone around the 45° line because entry rewards balanced core and support skills. When GenAI supplies the support task, the lower boundary for core specialists is replaced by the vertical threshold $\widehat{x}_i$ in (1), which passes through the apex of the pre-GenAI cone. GenAI supplies exactly the support level needed to match each founder's core skill. The added entry region therefore consists of individuals with strong core skills and weak support skills; the upper boundary is unchanged because support-dominant individuals do not benefit from adoption. Allowing the venture to hire and raise outside funds, as in Section 2.3, widens the added entry region further.

Figure A.2
GenAI Exposure Distribution at the Industry and Startup Levels



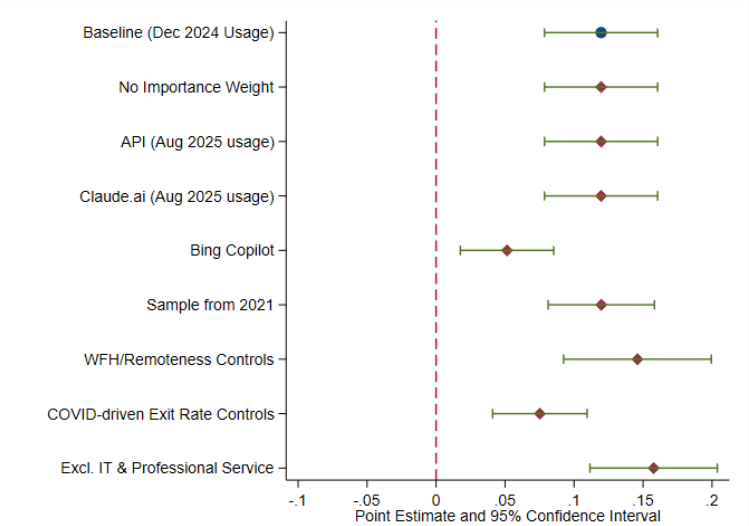
(a) Industry (four-digit NAICS)



(b) Startup

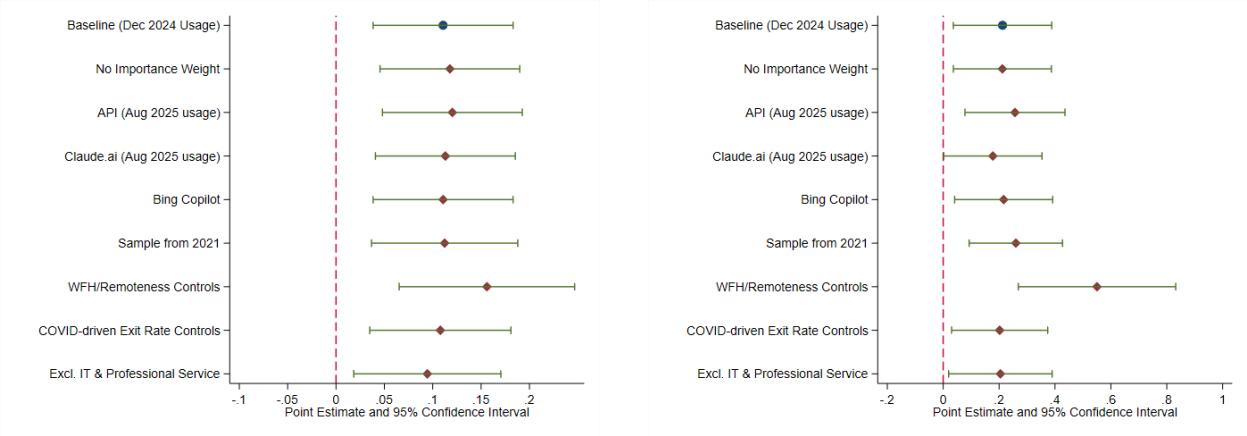
This figure illustrates the distribution of exposure to GenAI based on the Claude.ai conversation data. Panel (a) presents the histogram and kernel density of GenAI exposure aggregated at the four-digit NAICS industry level, while Panel (b) displays the corresponding distribution across startups.

Figure A.3
Effects of GenAI Introduction on Business Formation—Robustness



This figure presents robustness tests for the baseline estimates of the effect of GenAI introduction on corporate business applications, using the specification in Column (3) of Table 2. First, our conclusions are not driven by using interactions with Claude: we construct an alternative measure of GenAI exposure using Microsoft's *AI Applicability Index*, which not only captures how frequently Copilot is used for a given work activity, but also measures how effectively Copilot completes or assists with that activity and the share of the occupation's work that the activity represents. (Unlike the AEI, which measures the share of user conversations or traffic across O*NET tasks, Microsoft's index covers 330 O*NET intermediate work activities; these can be aggregated to the occupation level using the occupational activity weights, based on O*NET intermediate work activities.) The point estimates from the GenAI exposure measure based on Microsoft Bing Copilot are quantitatively and qualitatively similar to those from the AEI. Second, estimates are equally invariant if we define GenAI exposure using the September 2025 version of the AEI, which is based on August 2025 conversations. Finally, we consider that the diffusion of GenAI technologies largely coincides with the recovery from the COVID-19 pandemic and an increasing propensity to work from home: the estimated GenAI effects remain unchanged when we control for the industry exit rate during the 2020–2021 period, computed from the U.S. Census Bureau's Business Dynamics Statistics (BDS), and the industry propensity to work from home, obtained from Dingel and Neiman (2020). Furthermore, estimates are, if anything, stronger when we exclude 2020 from the estimation sample. The results also hold when we exclude the information technology and professional services sectors, indicating that they are not driven by the industries where GenAI adoption is most concentrated.

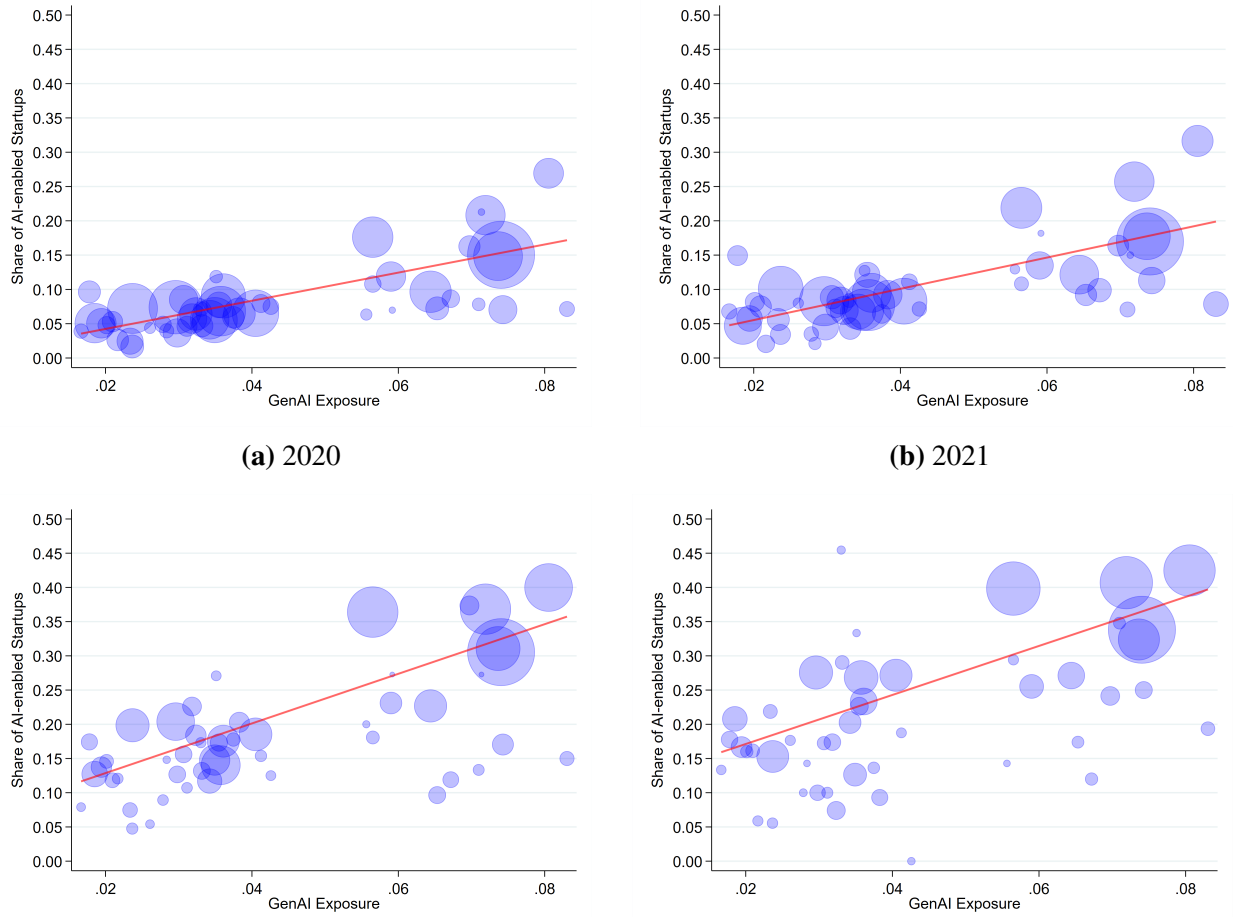
Figure A.4
Effects of GenAI Introduction on Firm Entry—Robustness



(a) Probability of Firm Entry **(b) # Firm Entry (log)**

This figure presents robustness tests for the baseline estimates of the effect of GenAI introduction on firm entry. Panel (a) reports the estimated effect on the probability of firm entry, and Panel (b) reports the effect on the (log) number of new firm entries. The specification in Panel (a) is analogous to Column (2) in Table 3, and the specification in Panel (b) corresponds to Column (4) in Table 3. First, our conclusions are not driven by using interactions with Claude: we construct an alternative measure of GenAI exposure using Microsoft's *AI Applicability Index*, which not only captures how frequently Copilot is used for a given work activity, but also measures how effectively Copilot completes or assists with that activity and the share of the occupation's work that the activity represents. (Unlike the AEI, which measures the share of user conversations or traffic across O*NET tasks, Microsoft's index covers 330 O*NET intermediate work activities; these can be aggregated to the occupation level using the occupational activity weights, based on O*NET intermediate work activities.) The point estimates from the GenAI exposure measure based on Microsoft Bing Copilot are quantitatively and qualitatively similar to those from the AEI. Second, estimates are equally invariant if we define GenAI exposure using the September 2025 version of the AEI, which is based on August 2025 conversations. Finally, we consider that the diffusion of GenAI technologies largely coincides with the recovery from the COVID-19 pandemic and an increasing propensity to work from home: the estimated GenAI effects remain unchanged when we control for the industry exit rate during the 2020–2021 period, computed from the U.S. Census Bureau's Business Dynamics Statistics (BDS), and the industry propensity to work from home, obtained from Dingel and Neiman (2020). Furthermore, estimates are, if anything, stronger when we exclude 2020 from the estimation sample. The results also hold when we exclude the information technology and professional services sectors, indicating that they are not driven by the industries where GenAI adoption is most concentrated.

Figure A.5
AI-Enabled Startups and Exposure to GenAI

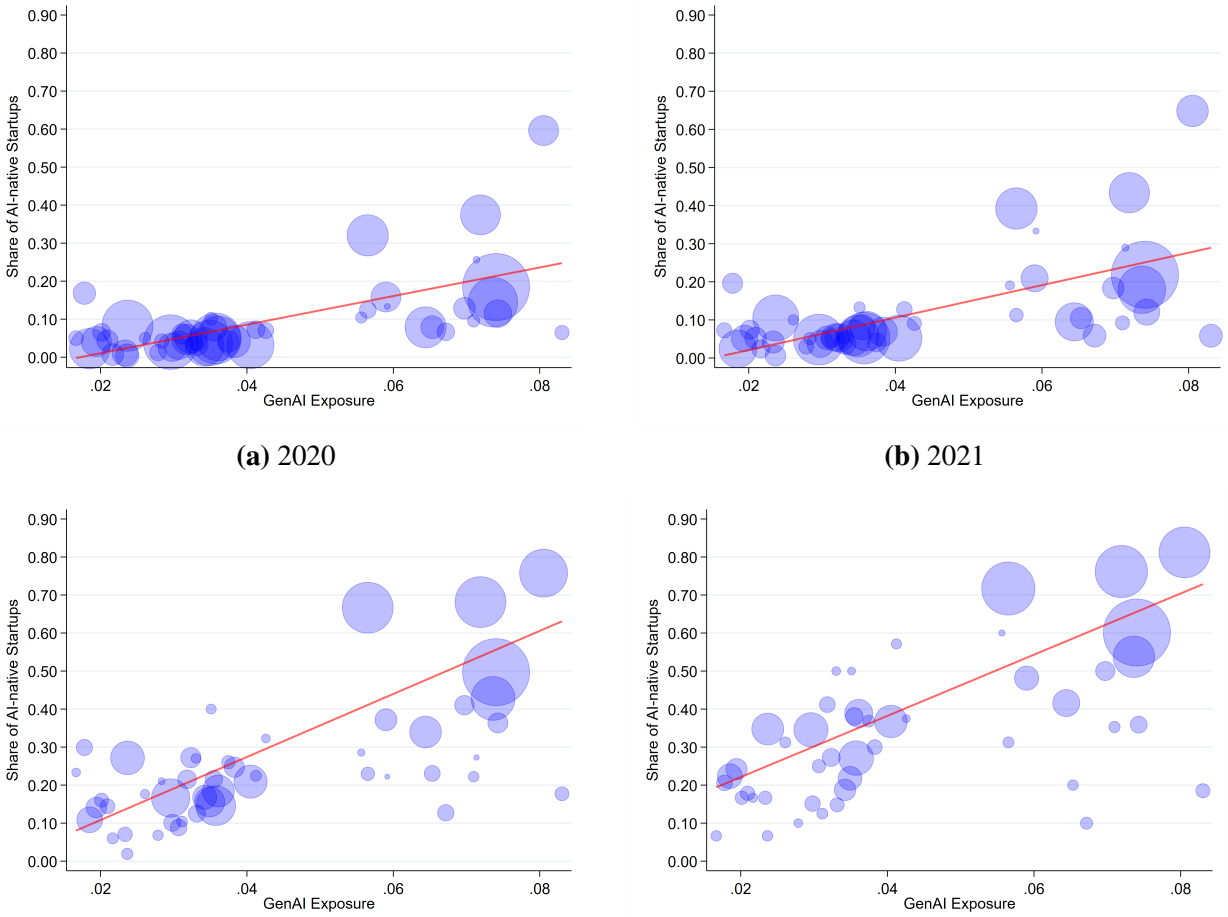


(a) 2020 **(b)** 2021

(c) 2023 **(d)** 2024

This figure plots the share of AI-enabled startups (identified using deep research) in each category group against the group's exposure to GenAI for the years before (2020-2021) and after (2023-2024) the introduction of GenAI. Each point represents a category group, with bubble size proportional to the total number of startups in that group in the corresponding year.

Figure A.6
AI-Native Startups and Exposure to GenAI

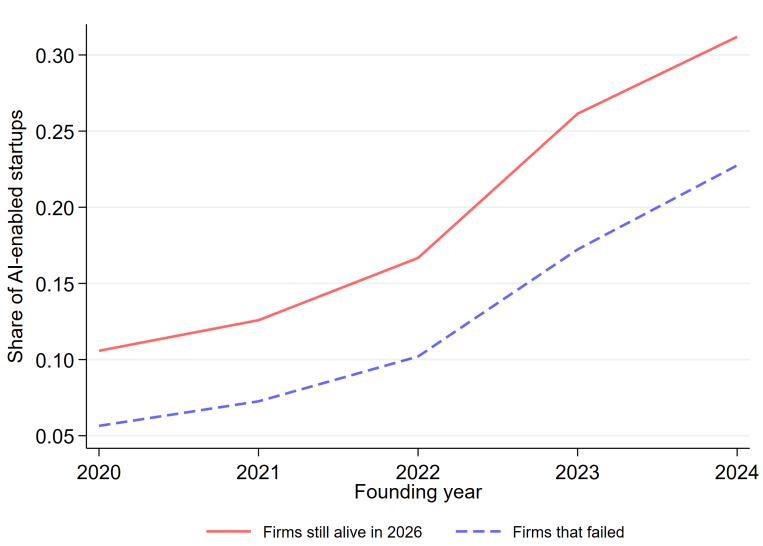


(a) 2020 **(b)** 2021

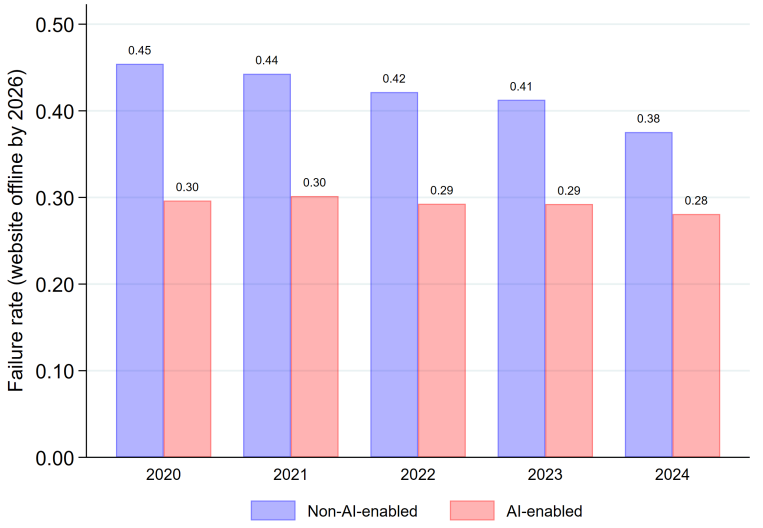
(c) 2023 **(d)** 2024

This figure plots the share of AI-native startups (identified using deep research) in each category group against the group's exposure to GenAI for the years before (2020–2021) and after (2023–2024) the introduction of GenAI. Each point represents a category group, with bubble size proportional to the total number of startups in that group in the corresponding year.

Figure A.7
Survival of AI-Enabled Startups



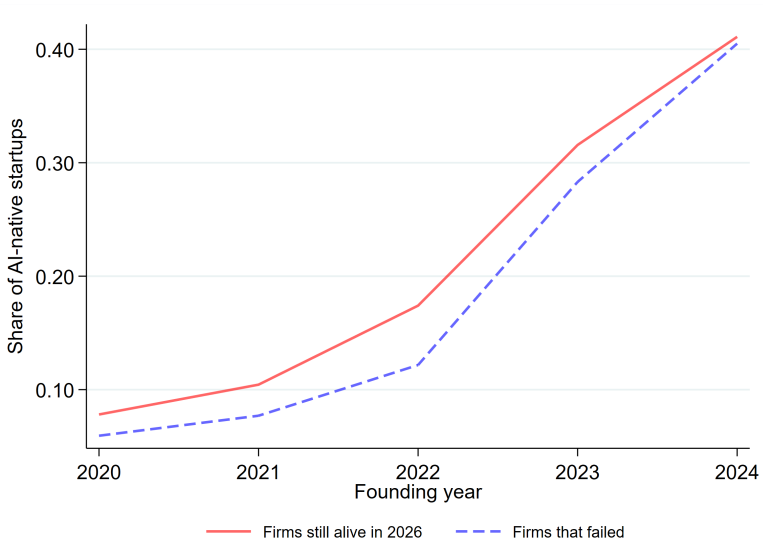
(a) Share of AI-Enabled Firms, Alive vs. Failed



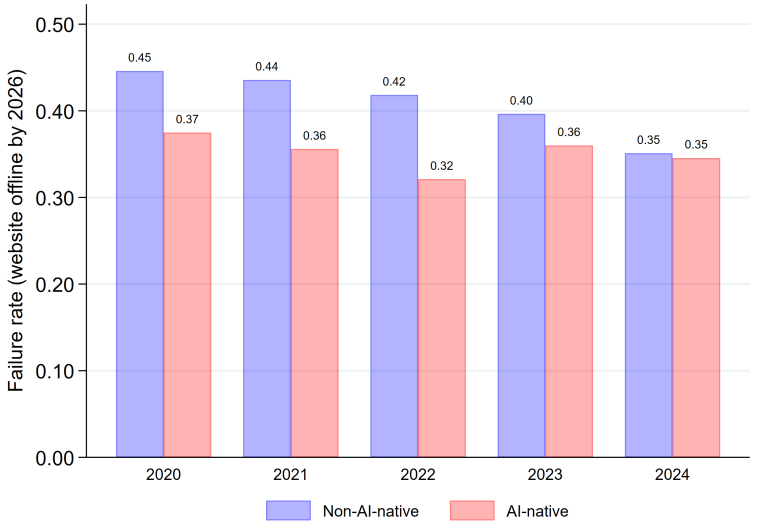
(b) Failure Rate, AI-Enabled vs. Non-AI-Enabled

This figure examines the relationship between AI-enabled status and startup survival. We define a startup as having failed if its website is offline by 2026. Panel (a) plots the share of AI-enabled firms among founding-year cohorts, separately for firms still alive in 2026 (solid line) and firms that failed (dashed line). Panel (b) plots the failure rate by founding year for AI-enabled versus non-AI-enabled startups.

Figure A.8
Survival of AI-Native Startups



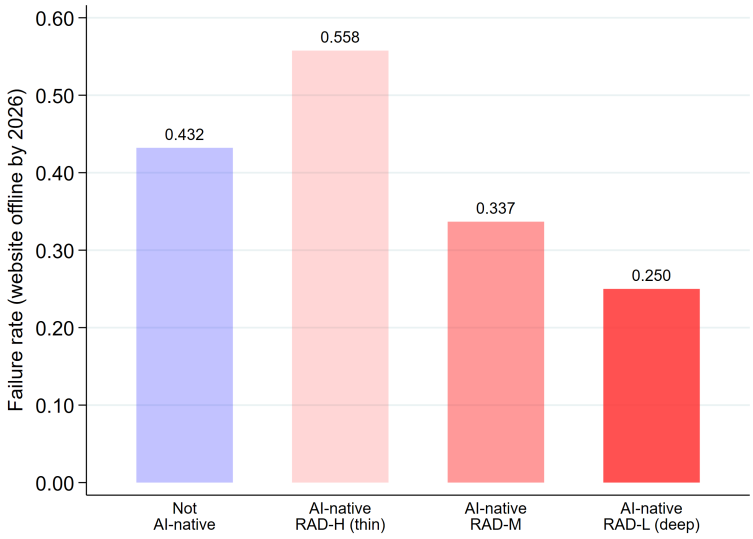
(a) Share of AI-Native Firms, Alive vs. Failed



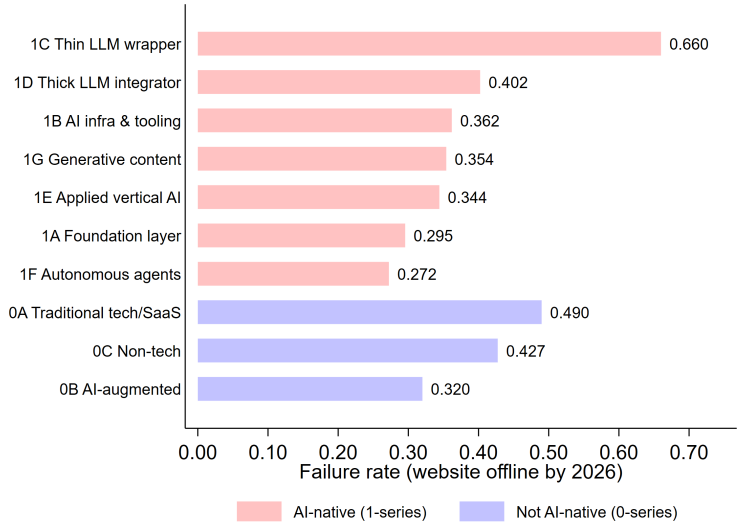
(b) Failure Rate, AI-Native vs. Non-AI-Native

This figure examines the relationship between AI-native status and startup survival. We define a startup as having failed if its website is offline by 2026. Panel (a) plots the share of AI-native firms among founding-year cohorts, separately for firms still alive in 2026 (solid line) and firms that failed (dashed line). Panel (b) plots the failure rate by founding year for AI-native versus non-AI-native startups.

Figure A.9
Failure Rates by AI-Reliance Depth and Industry Subclass



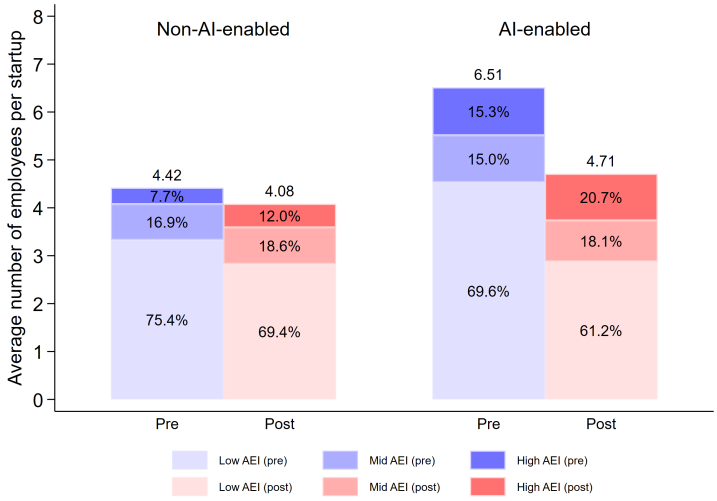
(a) By AI-Reliance Depth



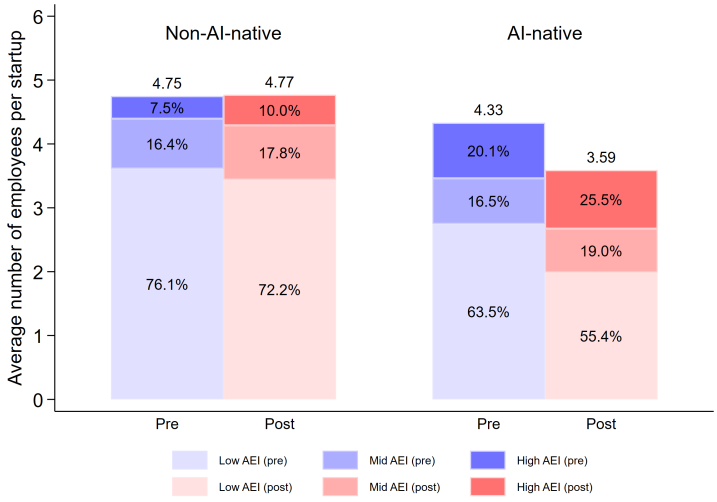
(b) By Industry Subclass

This figure examines failure rates (website offline by 2026) by AI-native status. Panel (a) plots the failure rate of non-AI-native startups, which serve as the baseline, and of AI-native startups at each level of reliance on third-party models or APIs: high structural dependence (RAD-H), medium dependence with some proprietary component (RAD-M), and low dependence with substantial in-house model development (RAD-L). Panel (b) plots failure rates by product subclass, separately for AI-native subclasses (1A–1G) and non-AI-native subclasses (0A–0C), ordered by failure rate within each group.

Figure A.10
Employee Composition of AI-Enabled and AI-Native Startups



(a) AI-Enabled Startups



(b) AI-Native Startups

This figure plots the average number of employees per startup at the end of the founding year—before and after the introduction of GenAI, separately by AI-enabled and AI-native classification. Panel (a) compares AI-enabled startups with non-AI-enabled startups. Panel (b) compares AI-native startups with non-AI-native startups. Each stacked bar reports the average number of employees per startup, decomposed into low-, mid-, and high-AEI employee groups. The numbers above the bars report the total average number of employees per startup. The percentages inside each segment report the share of each AEI group in total employment within the corresponding startup group and period.

B Appendix Tables

Table B.1
Summary Statistics at Industry-Year-Quarter Level (QWI)

Variable	N	Mean	Std. Dev.	Median	p1	p25	p75	p99
Employment (age=0-1)	5,218	18,434	67,520	3,893	22	1,006	12,491	140,538
Employment (age>1)	20,890	114,898	406,173	11,158	29	2,292	68,237	1,735,776
Earnings (age=0-1)	5,211	4,624	2,363	4,158	1,463	3,019	5,647	15,104
Earnings (age>1)	20,880	5,249	2,736	4,600	1,660	3,387	6,312	15,282

This table presents summary statistics on employment and earnings at the industry–year–quarter level, separately by firm age group, using data from the Census Bureau's Quarterly Workforce Indicators (QWI). The unit of observation is a four-digit NAICS industry–year–quarter cell within a firm age group, covering 2020Q1–2024Q4. *Employment* is the count of workers with stable jobs in the cell, defined as jobs held on both the first and last day of the quarter. *Earnings* is average monthly earnings per worker with stable employment. Firm age groups follow the QWI classification, with age measured from the date the firm first reports positive employment: *age=0-1* covers newly established firms, and *age>1* covers firms aged 2–3, 4–5, 6–10, and 11 or more years. Cells suppressed by the Census Bureau's disclosure avoidance procedures are excluded, so the number of observations varies across age groups.

Table B.2
Examples of O*NET Tasks by Share of Claude.ai Conversations

Task Description	Share (%)
Top 5 Tasks (Highest Shares)	
- Modify existing software to correct errors, to adapt it to new hardware, or to upgrade interfaces and improve performance.	6.65
- Diagnose, troubleshoot, and resolve hardware, software, or other network and system problems, and replace defective components when necessary.	2.82
- Modify existing software to correct errors, allow it to adapt to new hardware, or to improve its performance.	2.68
- Correct errors by making appropriate changes and rechecking the program to ensure that the desired results are produced.	2.46
- Write, update, and maintain computer programs or software packages to handle specific jobs such as tracking inventory, storing or retrieving data, or controlling other equipment.	2.40
Medium-Share Tasks	
- Advise managers on organizational policy matters such as equal employment opportunity and sexual harassment, and recommend needed changes.	0.006
- Collect information about individuals or clients, using interviews, case histories, observational techniques, and other assessment methods.	0.006
- Confer with engineers, customers, vendors, or others to discuss existing or potential electronics engineering projects or products.	0.006
- Develop and write procedures for installation, use, or troubleshooting of communications hardware or software.	0.006
- Identify or classify species of insects or allied forms, such as mites or spiders.	0.006
Low-Share Tasks	
- Teach fitness classes to improve the strength, flexibility, cardiovascular conditioning, or general fitness of participants.	0.002
- Test equipment performance, focus of lens system, diaphragm alignment, lens mounts, or film transport, using precision gauges.	0.002
- Treat wounds or superficial lacerations.	0.002
- Verify that transportation and handling procedures meet regulatory requirements.	0.002
- Write technical reports and draw charts that display statistics and data.	0.002

Table B.3
Examples of High and Low GenAI Exposure Occupations (8-digit O*NET-SOC code)

Occupation Title	Exposure
High Exposure Occupations	
Computer Programmers	0.5883
Software Developers	0.4689
Software Quality Assurance Analysts and Testers	0.2869
Network and Computer Systems Administrators	0.1744
Web Developers	0.1472
Web and Digital Interface Designers	0.1472
Tutors	0.1243
Technical Writers	0.1007
Computer Systems Analysts	0.0821
Writers and Authors	0.0808
Medium Exposure Occupations	
Social and Human Service Assistants	0.0007
Farmers, Ranchers, and Other Agricultural Managers	0.0007
Transportation, Storage, and Distribution Managers	0.0006
Exercise Trainers and Group Fitness Instructors	0.0006
Private Detectives and Investigators	0.0006
Cooks, Private Household	0.0006
Environmental Engineers	0.0006
Dental Hygienists	0.0006
Industrial Engineering Technologists and Technicians	0.0006
Shipping, Receiving, and Inventory Clerks	0.0006
Low Exposure Occupations	
Laundry and Dry-Cleaning Workers	0.0001
Aircraft Structure, Surfaces, Rigging, and Systems Assemblers	0.0001
Musical Instrument Repairers and Tuners	0.0001
Roofers	0.0001
Print Binding and Finishing Workers	0.0001
Heat Treating Equipment Setters, Operators, and Tenders, Metal and Plastic	0.0001
Welders, Cutters, Solderers, and Brazers	0.0001
Commercial Pilots	0.0001
Airline Pilots, Copilots, and Flight Engineers	0.0000
Architectural and Civil Drafters	0.0000

Table B.4
Examples of High and Low GenAI Exposure Industries (four-digit NAICS Level)

NAICS	Industry Description	Exposure
High Exposure Industries		
5132	Software Publishers	0.1207
5415	Computer Systems Design and Related Services	0.1135
3341	Computer and Peripheral Equipment Manufacturing	0.0626
3342	Communications Equipment Manufacturing	0.0520
3346	Manufacturing and Reproducing Magnetic and Optical Media	0.0455
3345	Navigational, Measuring, Electromedical, and Control Instruments Manufacturing	0.0397
4853	Taxi and Limousine Service	0.0375
5171	Wired Telecommunications Carriers	0.0368
5182	Data Processing, Hosting, and Related Services	0.0368
5192	Web Search Portals and All Other Information Services	0.0368
Medium Exposure Industries		
4411	Automobile Dealers	0.0039
5612	Facilities Support Services	0.0039
6213	Offices of Other Health Practitioners	0.0038
7212	RV (Recreational Vehicle) Parks and Recreational Camps	0.0037
4889	Other Support Activities for Transportation	0.0037
4572	Gasoline Stations with Convenience Stores	0.0037
6219	Other Ambulatory Health Care Services	0.0036
6211	Offices of Physicians	0.0034
3323	Architectural and Structural Metals Manufacturing	0.0034
3329	Other Fabricated Metal Product Manufacturing	0.0034
Low Exposure Industries		
6231	Nursing Care Facilities (Skilled Nursing Facilities)	0.0012
4922	Local Messengers and Local Delivery	0.0011
4883	Support Activities for Water Transportation	0.0011
4931	Warehousing and Storage	0.0010
4911	Postal Service	0.0010
4852	Interurban and Rural Bus Transportation	0.0010
2122	Metal Ore Mining	0.0010
4921	Couriers and Express Delivery Services	0.0009
3116	Animal Slaughtering and Processing	0.0009
2121	Coal Mining	0.0008

Table B.5
GenAI Exposure by Crunchbase Category Group

Category Group	Exposure	Category Group	Exposure
Blockchain and Cryptocurrency	0.0830	Commerce and Shopping	0.0342
AI	0.0805	Lending and Investment	0.0332
Apps	0.0743	Travel and Tourism	0.0330
Software	0.0740	Education	0.0323
Information Technology	0.0736	Administrative Services	0.0318
Data and Analytics	0.0719	Sports	0.0311
Navigation and Mapping	0.0714	Advertising	0.0307
Gaming	0.0710	Community and Lifestyle	0.0298
Privacy and Security	0.0697	Professional Services	0.0296
Payments	0.0672	Clothing and Apparel	0.0284
Mobile	0.0653	Events	0.0278
Internet Services	0.0644	Agriculture and Farming	0.0261
Platforms	0.0592	Health Care	0.0237
Hardware	0.0590	Social Impact	0.0237
Consumer Electronics	0.0565	Food and Beverage	0.0234
Science and Engineering	0.0565	Consumer Goods	0.0217
Messaging and Telecommunication	0.0556	Sustainability	0.0209
Music and Audio	0.0426	Energy	0.0201
Video	0.0412	Manufacturing	0.0194
Other	0.0405	Real Estate	0.0185
Design	0.0383	Biotechnology	0.0178
Content and Publishing	0.0375	Natural Resources	0.0167
Sales and Marketing	0.0361		
Financial Services	0.0357		
Transportation	0.0355		
Government and Military	0.0351		
Media and Entertainment	0.0349		

Table B.6
Top Ranked Startups based on GenAI Exposure

Company	Short Description
SuperDuperAi	SaaS for Generative AI.
Sapix Lab, Inc.	Redefines entertainment products through the power of Generative AI.
CRACKENAGI, Inc.	IT system testing and evaluation.
Comfy Deploy, Inc.	SaaS company providing production-ready APIs for generative AI applications.
Experio Labs, Inc.	Knowledge management platform using AI and machine learning.
Ablt AI, Inc.	Enables rapid deployment of Generative AI capabilities for any business.
Cloobot Techlabs Pvt Ltd.	Builds implementation-as-a-service solutions for enterprise CRM/ERP systems.
Langgenius, Inc.	Dify is an open-source LLM app development platform integrating Backend-as-a-Service and LLMOps.
Infinitform, Inc.	AI-powered design tool optimizing CAD parts for real-world manufacturing processes.
Lilac Voice AI	Conversational AI agent for drive-thru order taking.
heyLevi	Zero-prompt AI software that creates brand identity and positioning for businesses.
TalkStack AI	No-code GenAI customer experience platform for enterprises without AI teams.
Simprobot, Inc.	On-premise AI solution for corporations.
Statra LLC	Analytics platform addressing reporting challenges in healthcare organizations.
LOMA (US)	Marketing platform to scale and validate local marketing for multi-unit brands.
Tabnam, Inc.	Greptile provides AI-driven code review and development tool integration.
Preloop, Inc.	Platform to deploy ML models in hours instead of weeks.
Anote	Human-centered AI platform combining generative models with human feedback on edge cases.
Goml	Enterprise solutions provider in data engineering, MLOps, and LLM applications.
Pull Data, Inc.	Machine-learning-as-a-service solution for capturing and analyzing vehicle data.

Table B.7

GenAI Introduction and Business Formation: Alternative Driscoll–Kraay Bandwidths

Panel A: Bandwidth of six lags

	(1)	(2)	(3)	(4)	(5)
Dep. Var.	BA	HBA	CBA	PBF4Q	PBF8Q
AEI (above median) × Post	0.020	0.083***	0.120***	0.075**	0.081**
	[0.023]	[0.024]	[0.037]	[0.029]	[0.032]
Sector FE	YES	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES	YES
Obs	1,105	1,105	1,105	1,099	1,099
Adj R2	0.993	0.993	0.988	0.988	0.984

Panel B: Bandwidth of twelve lags

	(1)	(2)	(3)	(4)	(5)
Dep. Var.	BA	HBA	CBA	PBF4Q	PBF8Q
AEI (above median) × Post	0.020	0.083***	0.120***	0.075**	0.081**
	[0.026]	[0.028]	[0.042]	[0.029]	[0.032]
Sector FE	YES	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES	YES
Obs	1,105	1,105	1,105	1,099	1,099
Adj R2	0.993	0.993	0.988	0.988	0.984

This table reports the estimated effects of GenAI introduction on new business formation using aggregate-level data from the U.S. Census Bureau's *Business Formation Statistics* (BFS), reproducing Table 2 with wider Driscoll–Kraay bandwidths. Panel A uses six monthly lags and Panel B uses twelve; the main table uses four. The outcome variable is the logarithm of the number of new business formations in a two-digit NAICS industry, measured monthly. The sample covers January 2020 to May 2025. AEI (above median) is a dummy that equals one for sectors whose AI exposure, measured by the Anthropic Economic Index (AEI) aggregated to the two-digit NAICS level, is above median, and zero otherwise. *Post* is a dummy that equals one from August 2022 onward and zero otherwise. Columns (1)-(5) use five BFS outcome measures: (1) all Business Applications (BA), (2) High-Propensity Business Applications (HBA), (3) Business Applications from Corporations (CBA), (4) Projected Business Formations within four quarters (PBF4Q), and (5) Projected Business Formations within eight quarters (PBF8Q). HBA represent EIN applications with characteristics historically associated with the successful establishment of active firms, such as intended payroll or corporate registration. PBF measures are Census-derived projections of business applications expected to become active employer firms within four to eight quarters and are considered a leading indicator of realized business creation. Sector (two-digit NAICS) and month-year fixed effects are included in all specifications. Standard errors are Driscoll–Kraay, which are robust to correlation across sectors within a month and to serial correlation within a sector. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table B.8
GenAI Introduction and Firm Entry: Count Models

	(1)	(2)
Dep. Var.	# firm entry (count)	
	Negative binomial	Poisson (PPML)
AEI (above median) × Post	0.357***	0.188*
	[0.131]	[0.098]
Implied effect (%)	42.8	20.7
Sample	Full	# entry $\leq$ p90
Industry FE	YES	YES
Year FE	YES	YES
Obs	1,430	1,249

This table presents a robustness check for Column (4) of Table 3, which estimates the effect of GenAI introduction on the number of new firm entries. Column (1) estimates a negative binomial model on the full sample. Column (2) estimates a Poisson pseudo-maximum-likelihood (PPML) model. Because the PPML score weights each observation by its fitted mean, industry–year cells with very large entry counts dominate the full-sample estimate; Column (2) therefore excludes cells above the 90th percentile of the entry distribution. Coefficients from both models are semi-elasticities, so the implied proportional effect is $\exp(\beta) - 1$; this is reported in the row labeled *Implied effect*. The unit of observation is a four-digit NAICS industry–year cell, and the sample includes newly established, single-location firms founded between 2020 and 2024. *Post* is a dummy that equals one for 2023–2024 and zero otherwise. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year fixed effects are included in both specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table B.9

Channels Behind Dispersed Entry: Labor Specialization and Financing

Alternative Measures of Labor Specialization and VC hubs

Panel A: Relaxed Labor Specialization Requirements

	(1)	(2)	(3)	(4)
Dep. Var.	I (firm entry)		# firm entry	
	non-specialized	specialized	non-specialized	specialized
AEI (above median) × Post	0.091**	0.046	0.204**	0.055
	[0.037]	[0.037]	[0.088]	[0.061]
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430
Adj R2	0.477	0.574	0.887	0.851

Panel B: Relaxed Financing Requirements

	(1)	(2)	(3)	(4)
Dep. Var.	I (firm entry)		# firm entry	
	non VC hubs	VC hubs	non VC hubs	VC hubs
AEI (above median) × Post	0.107***	-0.01	0.216**	-0.035
	[0.036]	[0.038]	[0.087]	[0.063]
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430
Adj R2	0.491	0.58	0.89	0.81

This table examines two channels through which the introduction of GenAI may have contributed to more geographically dispersed firm entry with alternative measures of labor specialization and VC hubs, using establishment-level data aggregated from the InfoGroup firm panel. The unit of observation is a four-digit NAICS industry–year cell. The sample includes newly established, single-location firms founded between 2020 and 2024. Panel A examines the labor specialization channel using a broader geographic unit. Each combined statistical area (CSA) is classified as having *specialized* or *non-specialized* labor in a given industry based on the pre-GenAI distribution of employment. For each industry, the share of local employment in that industry relative to total CSA employment is computed, and CSAs in the top decile of this distribution are defined as specialized. Panel B tests the financing channel using an alternative definition of VC hubs. Venture capital intensity is measured using data from Crunchbase on all U.S.-based investors. The total dollar value of investments made by investors located in each county is aggregated, and the top ten counties in this distribution are classified as VC hubs, with all others treated as non-hubs. The dependent variable is an indicator for any firm entry, *I(firm entry)*, in Columns (1) and (2), and the logarithm of one plus the number of new firm entries, *# firm entry*, in Columns (3) and (4), each measured at the industry–year cell level and computed separately for entry in non-specialized (Columns (1) and (3)) or specialized (Columns (2) and (4)) CSAs in Panel A, and in non-hub (Columns (1) and (3)) or hub (Columns (2) and (4)) counties in Panel B. *Post* is a dummy that equals one for 2023–2024 and zero otherwise. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year fixed effects are included in all specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table B.10
GenAI Introduction and the Size Composition of New Firms
Absolute Size Thresholds

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	# firm entry					
	$\leq$ 3 employees	$>$ 3 employees	$\leq$ 5 employees	$>$ 5 employees	$\leq$ 10 employees	$>$ 10 employees
AEI (above median) $\times$ Post	0.183**	0.031	0.174**	0.029	0.180**	0.06
	[0.075]	[0.087]	[0.079]	[0.085]	[0.082]	[0.083]
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Obs	1,430	1,430	1,430	1,430	1,430	1,430
Adj R2	0.873	0.877	0.876	0.853	0.877	0.769

This table reports the estimated effects of GenAI introduction on the size distribution of new firm entries, using establishment-level data aggregated from the *InfoGroup* firm panel. The unit of observation is a four-digit NAICS industry–year cell. The sample includes newly established, single-location firms founded between 2020 and 2024. The dependent variable is the logarithm of one plus the number of new firm entries (*# firm entry*) in each industry–year cell, computed separately for firms with initial employment at or below versus above an absolute size threshold of 3 employees (Columns (1) and (2)), 5 employees (Columns (3) and (4)), and 10 employees (Columns (5) and (6)). *Post* is a dummy that equals one for 2023–2024 and zero otherwise. *AEI (above median)* is a dummy equal to one for industries whose exposure is above median, and zero otherwise. Industry (four-digit NAICS) and year fixed effects are included in all specifications. Standard errors are clustered at four-digit NAICS level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table B.11

GenAI Introduction and Startup Founders' Prior Industry Experience (four-digit NAICS)

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	≥3 Years in Startup Industry	Ever in Startup Industry	Share of Years in Startup Industry	≥3 Years in Startup Industry	Ever in Startup Industry	Share of Years in Startup Industry
AEI (above median) × Post	0.067**	0.058**	0.031*	0.074***	0.068**	0.038**
	[0.026]	[0.029]	[0.017]	[0.026]	[0.028]	[0.017]
Dep. Var. Mean	0.253	0.444	0.196	0.253	0.444	0.196
Industry Category Indicators	NO	NO	NO	YES	YES	YES
Year FE	NO	NO	NO	YES	YES	YES
Obs	15,236	15,236	14,407	15,236	15,236	14,407
Adj R2	0.004	0.007	0.001	0.019	0.020	0.009

This table reports the estimated effects of GenAI introduction on founders' prior industry experience, defined at the four-digit NAICS level, using founder-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is an individual founder of startups founded between 2020 and 2024. The dependent variables measure the founder's prior work experience in the startup's own four-digit NAICS industry: an indicator for at least three years of prior experience (*≥3 Years in Startup Industry*); an indicator for any prior experience (*Ever in Startup Industry*); and the share of the founder's career spent in that industry (*Share of Years in Startup Industry*). *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. Columns (1)–(3) report estimates without startup category-group controls or year fixed effects, while Columns (4)–(6) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table B.12
GenAI Introduction and Startup Founders' Occupational Backgrounds

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	AEI of past positions					
	Most Recent	Mean	Max	Most Recent	Mean	Max
AEI (above median) × Post	0.021***	0.014***	0.027***	0.017***	0.009*	0.019**
	[0.005]	[0.004]	[0.008]	[0.006]	[0.005]	[0.008]
Dep. Var. Mean	0.047	0.047	0.092	0.047	0.047	0.092
Industry Category Indicators	NO	NO	NO	YES	YES	YES
Year FE	NO	NO	NO	YES	YES	YES
Obs	14,787	14,787	14,787	14,787	14,787	14,787
Adj R2	0.048	0.065	0.081	0.069	0.097	0.121

This table examines how the occupational backgrounds of startup founders differ across industries with varying exposure to GenAI, using founder-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is an individual founder of startups founded between 2020 and 2024. The dependent variables measure how exposed each founder's pre-founding career was to GenAI-related technological change. Specifically, Columns (1) and (4) use the AEI of the founder's most recent occupation prior to founding, Columns (2) and (5) use the mean AEI across all pre-founding occupations, and Columns (3) and (6) use the maximum AEI observed across the founder's career history. Higher values indicate that the founder's prior jobs were more susceptible to GenAI-related automation or transformation. *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. Columns (1)–(3) report estimates without startup category-group controls or year fixed effects, while Columns (4)–(6) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table B.13
GenAI Introduction and Startup Founder Traits: Observed Traits Only

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dep. Var.	Capital/ Investing	AI/ML GenAI	Product Manager/ Development	Marketing/ Sales	Operations/ Logistics	Capital/ Investing	AI/ML GenAI	Product Manager/ Development	Marketing/ Sales	Operations/ Logistics
AEI (above median) × Post	-0.156**	0.239***	-0.089**	-0.065	-0.037	-0.079	0.081**	-0.104**	-0.024	-0.009
	[0.072]	[0.044]	[0.039]	[0.088]	[0.073]	[0.059]	[0.031]	[0.040]	[0.073]	[0.064]
Dep. Var. Mean	0.425	0.217	0.386	0.700	1.096	0.425	0.217	0.386	0.700	1.096
Industry Category Indicators	NO	NO	NO	NO	NO	YES	YES	YES	YES	YES
Year FE	NO	NO	NO	NO	NO	YES	YES	YES	YES	YES
Obs	15,154	15,154	15,154	15,154	15,154	15,154	15,154	15,154	15,154	15,154
Adj R2	0.060	0.082	0.155	0.183	0.359	0.161	0.176	0.171	0.270	0.388

This table reports the estimated effects of GenAI introduction on observed founder trait counts, using founder-level data from the Crunchbase–LinkedIn matched sample. The unit of observation is an individual founder of startups founded between 2020 and 2024. The dependent variables are counts of observed traits in five trait clusters defined in Appendix D: Cluster 3, capital, investing, and deal mechanics; Cluster 8, AI/ML and GenAI; Cluster 5, product building and "0–1" creation; Cluster 4, go-to-market, including marketing, sales, growth, and partnerships; and Cluster 12, operations, service delivery, and logistics. *Post* is a dummy that equals one for startups founded from August 2022 onward and zero otherwise. *AEI (above median)* is a dummy equal to one for firms operating in category groups with above-median GenAI exposure, and zero otherwise. All columns control for the total number of observed traits extracted for the founder across the 16 trait groups. Columns (1)–(5) report estimates without startup category-group controls or year fixed effects, while Columns (6)–(10) include both. Standard errors are clustered on 50 bins of startup GenAI exposure. Because a startup can belong to several Crunchbase category groups at once, there is no partition of firms on which to cluster at the category-group level; we therefore sort startups into 50 equal-sized bins of *AEI*, comparable to the number of distinct category groups in the sample, so that each firm belongs to exactly one cluster. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

C Identifying AI-Enabled and AI-Native Startups

We use LLM-based pipelines to construct two startup-level indicators from company records and public web evidence for the U.S. Crunchbase–Revelio matched sample described in Section 3. The sample contains 44,387 U.S. startups founded between 2020 and 2024 that have between one and 100 employees at the end of the founding year. The first indicator, `AI_enabled`, equals one when the pipeline verifies at least one citable public source explicitly documenting the startup's internal use of a generative-AI tool in a business function. The second indicator, `AI_native`, equals one when the pipeline determines from current or archived website evidence that AI or machine learning is essential to the startup's core product or service—that is, the product or service would fundamentally not exist without it. The AI-native construct is deliberately broader than generative AI and includes earlier applications of machine learning, such as computer vision, natural language processing, recommendation systems, and forecasting. The indicators measure different features and can both equal one: a firm may use GenAI internally, build an AI-native product, satisfy both definitions, or satisfy neither. Because internal operating practices and product offerings appear in different types of public evidence, we construct the two indicators with separate pipelines. The pipelines are designed to collect broad raw evidence and classify it precisely while keeping data-construction costs low enough to operate at scale.

C.1 Identifying AI-Enabled Startups

We identify AI-enabled startups using a three-stage, LLM-based agentic pipeline that converts public web evidence into a firm-level indicator. Stage 1 identifies firms with enough public information to research reliably, Stage 2 searches for citable evidence of internal GenAI use, and Stage 3 verifies each candidate finding against its cited source and creates the firm-level indicator. Figure C.1 summarizes the pipeline and the number of firms remaining after each stage.

Stage 1: Public-information screen. Stage 1 begins with each firm's name, homepage URL, and short and long Crunchbase descriptions when available. We record whether the homepage responds, but treat a nonresponding site only as a signal and never as an automatic reason for exclusion. We then make one call per company to the `Tavily Search` API, using the query "*Find information on the company {name}.*" Withholding the homepage and descriptions tests whether the firm can be identified unambiguously from public web results without directing the search toward its Crunchbase record.

On February 9, 2026, `GPT-5 nano` compared the Tavily results with the firm's Crunchbase profile and assigned a research-priority score from zero to five. Firms scoring four or five advanced to Stage 2. Of the 44,373 firms that received a score, 9,420 met this threshold, representing 21.2% of the 44,387-firm starting sample. The instructions used to assign this score are listed as Stage 1 in Table C.2.

Stage 2: Targeted evidence search. The second stage searches the public web for citable evidence that each of the 9,420 firms uses GenAI in its own operations. To accommodate the different ways in which such use becomes public, we search three evidence channels separately: (i) job postings; (ii) firm-owned sources, including company websites, blogs, documentation, and official accounts; and (iii) third-party sources, including news reports, interviews, podcasts, vendor case studies, and employee posts.

The search has two parts. First, three short screening searches, called "scouts," examine the evidence channels separately. We run a full search, called a "dig," for a channel only when its scout rates the available material as moderate or strong and returns at least one URL. A firm can therefore receive between zero and three full searches. Each dig begins a new search using only the firm's identity and the assigned evidence channel, rather than the scout's links. This design reserves the longer searches for promising channels without tying them to the scout's initial results. All scouts and digs were run through the `Perplexity Agent API` using `OpenAI GPT-5.6 Luna` in the production run conducted on August 16–17, 2026. Table C.1 reports their principal settings.

Table C.1
Agent configurations for the targeted evidence search

Parameter	Scout: initial screen (three per firm)	Dig: full search (zero to three per firm)
Task	Determine whether the assigned evidence channel contains promising public material	Determine whether that channel contains citable evidence of internal GenAI use
Model	`OpenAI GPT-5.6 Luna`	`OpenAI GPT-5.6 Luna`
Tools	Web search only	Web search and page retrieval
Maximum steps	2	50
Search depth	Low	Medium
Reasoning effort	Low	High

Note: A dig is launched only for a channel whose scout returns at least one URL and rates the available material as moderate or strong.

The decisive rule is *use, not sell*. Evidence that a startup sells an AI product does not, by itself, show that the startup uses GenAI in its own operations. The pipeline records a finding only when a source (i) explicitly describes internal use and (ii) supplies a citable URL. Findings are tool-specific whenever the source permits. For example, a generic label such as "AI coding tool" is not retained when the source names a product such as `GitHub Copilot`, `Cursor`, or `DALL-E`. If the source describes qualifying use without naming the product, the record states that the tool is unnamed. Each record also contains the business function and use case, an evidence summary, the URL and source type, and the evidence channel.

The agents receive a common set of instructions plus instructions specific to each evidence channel. The corresponding scout and dig files are listed as Stage 2 in Table C.2.

Stage 3: Removing duplicates, verifying sources, and classifying firms. We first use fixed rules, without an LLM, to remove duplicate or republished versions of the same evidence. Distinct tools or use cases cited by the same source are retained. This step reduces the 34,169 raw findings for 5,496 startups produced in Stage 2 to 32,880 distinct records while preserving all 5,496 startups.

We then reopen each cited URL without conducting a new search. The verifier first uses Perplexity's `fetch_url`, the same retrieval tool used in Stage 2. If it cannot retrieve the page, the verifier uses `Tavily Extract` (`https://api.tavily.com/extract`) as a backup and, if that also fails, attempts a direct HTTP request. The initial Perplexity retrieval failed for 1,999 records, which were then passed to Tavily Extract; 81 records required the direct HTTP request. If none of these methods returns usable text, the record remains unresolved.

If a retrieval method yields readable text, a separate verification agent—`OpenAI GPT-5.6 Luna`, executed on August 18, 2026—compares the reopened text with the saved evidence summary and returns a yes-or-no decision. The agent receives no new search results and may not rely on outside knowledge: the cited page must itself support the claim. A record is verified only when the readable page supports the claim. It is rejected when the page does not support the claim, including when the page documents the startup selling rather than using AI. If the page cannot be read, or if a technical error prevents a decision, the record is left unresolved rather than rejected. The verification instructions are listed as Stage 3 in Table C.2.

For each record that receives a decision, the system calculates confidence from the probabilities that the model assigns to the two possible answers: verified and rejected. It records the probability assigned to the answer the model selects, rather than asking the model to describe its confidence in words. This measure is available for all 30,062 records that receive a decision. Its mean is 98.4%, its median exceeds 99.99%, and 95.1% of decisions receive at least 95% probability. These values show that the verification agent usually favors one answer strongly, but they do not measure the actual probability that its decision is correct. As a robustness check, requiring at least one verified record with confidence of 95% or more would identify 4,971 startups, or 98.9% of the original 5,028. The firm-level classification is therefore not driven by the small number of lower-confidence decisions.

Among the 32,880 records remaining after duplicate removal, 28,535 (86.8%) are verified, 1,527 (4.6%) are rejected, and 2,818 (8.6%) remain unresolved because the cited page could not be checked. Among the 30,062 records that receive a decision, 94.9% are verified. We classify a startup as AI-enabled if it has at least one verified record. This procedure identifies 5,028 firms, corresponding to 53.4% of the 9,420 researched firms and 11.3% of the full sample.

Descriptive patterns. Verified evidence is distributed across third-party sources (38.0%), firm-owned media (33.9%), and job postings (28.2%); no single channel drives the measure. At the firm level, the most frequently documented functions are operations (50.4% of AI-enabled firms), engineering (45.2%), and marketing (33.5%).

The documented AI-enabled share rises from 47.4% among researched firms founded in 2020

to 66.1% among those founded in 2024. In the full sample, it rises from 7.8% to 26.3%. The latter change also reflects an increase in the share of firms passing the public-information screen, from 16.5% to 39.9% across the same cohorts.

For transparency, Table C.2 brings together the prompt files used for the LLM-based decisions in all three stages. The table lists the files but does not reproduce their contents.

Table C.2
Prompt files used in the AI-enabled classification pipeline

Stage	Prompt role	File
Stage 1	Public-information classifier	`stage_1_classifier.txt`
Stage 2	Scout: shared instructions	`scout_shared_preamble.txt`
Stage 2	Scout: job postings	`scout_jobs.txt`
Stage 2	Scout: firm-owned sources	`scout_owned.txt`
Stage 2	Scout: third-party sources	`scout_third_party.txt`
Stage 2	Dig: shared instructions	`dig_shared_preamble.txt`
Stage 2	Dig: job postings	`dig_jobs.txt`
Stage 2	Dig: firm-owned sources	`dig_owned.txt`
Stage 2	Dig: third-party sources	`dig_third_party.txt`
Stage 3	Source-verification agent	`judge.txt`

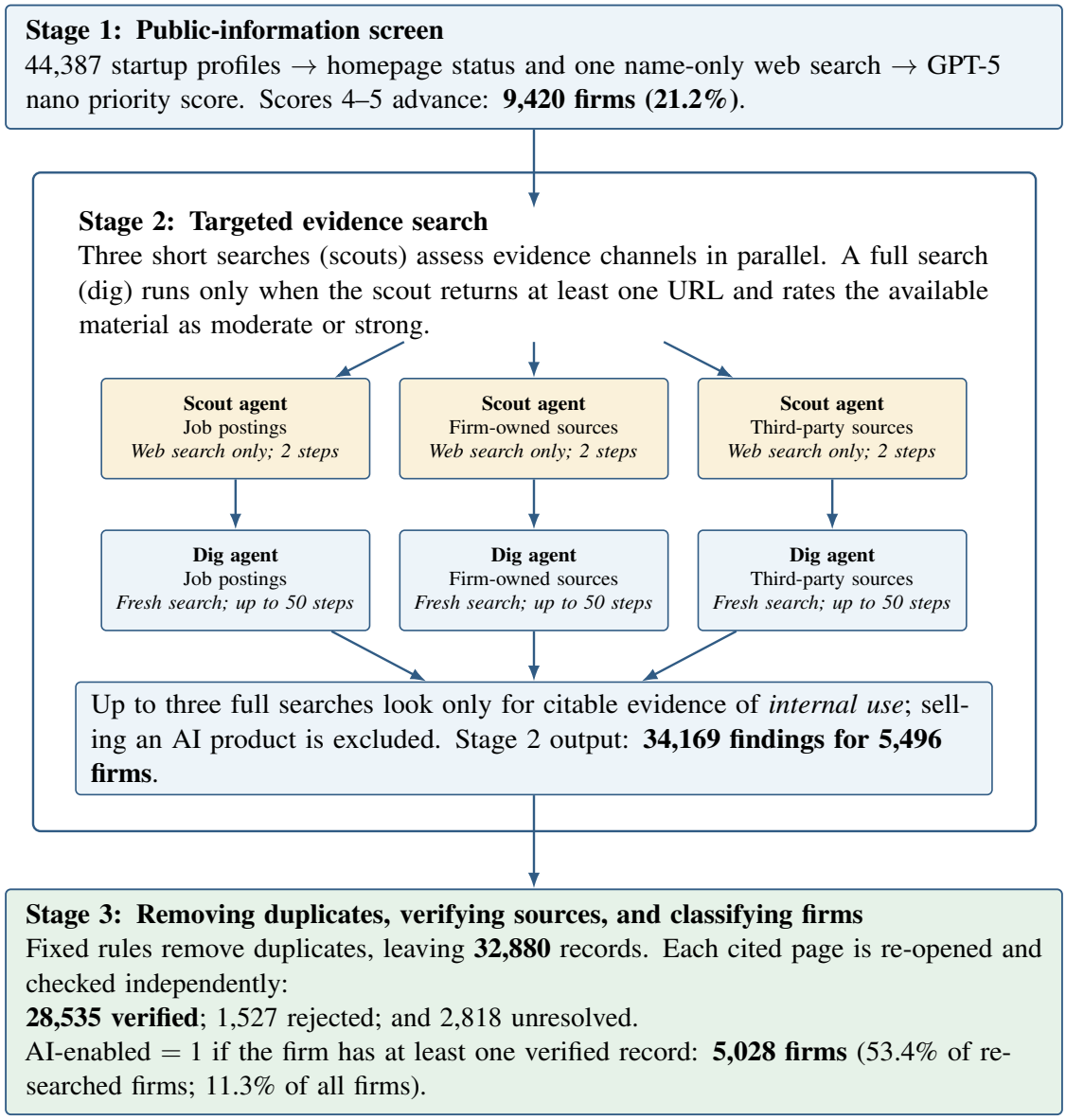


Figure C.1
Workflow for identifying AI-enabled startups. The classification requires independently verified public evidence of internal GenAI use. Counts before the final firm classification refer to evidence records; all other counts refer to firms.

C.2 Identifying AI-Native Startups

We identify AI-native startups using a three-stage, LLM-based classification pipeline that converts company information and website evidence into a firm-level indicator. Stage 1 collects evidence from current and archived websites, Stage 2 determines whether each firm is AI-native, and Stage 3 assigns the product category and reliance on third-party generative-AI infrastructure. Figure C.2 summarizes the pipeline and the number of firms reaching each stage.

Stage 1: Collecting website evidence and searching archives. Stage 1 begins with the same 44,387 matched startups. The company information supplied to the pipeline includes short and long descriptions, category keywords, founding date, employee count, and funding. These fields provide context, but website text is the primary evidence about what the firm builds and sells because platform descriptions can be stale or generic. We use `Tavily Crawl` (`https://api.tavily.com/crawl`) to retrieve text from one to five pages of each usable current website. The crawler receives the following instruction: "*Select up to five pages that best explain what this company does, sells, who it serves, and how the offering works. Prefer the homepage and product, platform, solutions, services, use-case, about, pricing, documentation, technical, or research pages containing concrete evidence.*"

Using only firms with a current website, however, would disproportionately retain firms whose websites remained online through the collection date and would omit firms that disappeared, rebranded, or lost a retrievable site.

For firms without usable current-site evidence, we therefore retrieve historical website snapshots from the Internet Archive's Wayback Machine (`https://web.archive.org/`). We treat the last available capture as an approximate date when the site disappeared, seek an earlier snapshot from approximately six months before that capture, and extract the selected archived homepage. For archived websites, we provide no page-selection instructions and use `Tavily Extract` (`https://api.tavily.com/extract`) to retrieve the selected homepage.

Of the 22,002 firms without current website evidence but with an address suitable for an archive search, 19,044 were located in the archive and 15,714 yielded usable website evidence. Combining these firms with the 22,032 firms that have current website evidence produces an evidence sample of 37,746 firms, or 85.0% of the starting sample. Archived evidence generally contains less text because it consists of one homepage rather than several current pages. The labels "current" and "archived" describe the source of the evidence; they do not establish whether the firm is legally operating, and the archive date is not a formal closure date. Stage 2 subsequently completed classification for 37,218 firms: 21,506 with current-site evidence and 15,712 with archived-site evidence. Because Stage 1 retrieves and assembles website evidence using code rather than an LLM, it has no prompt file.

Stage 2: AI-native classification. The production run used `OpenAI GPT-5.6 Luna` on July 28, 2026. The model determines whether AI or machine learning is essential to the firm's core product

rather than merely an added feature. Specifically, it asks whether the product or service would fundamentally still exist without AI or machine learning. We classify a firm as AI-native only when it would not. Making this decision separately from the detailed product categories allows the confidence measure to reflect the AI-native decision alone. The instructions used for this decision are listed as Stage 2 in Table C.4.

For each firm, the system calculates confidence from the probabilities that the model assigns to the two possible answers: AI-native and not AI-native. It records the probability assigned to the answer the model selects, rather than asking the model to describe its confidence in words. This measure is available for all 37,218 classified firms. Its mean is 99.6%, and 99.1% of decisions receive at least 90% probability. These values show that the model usually favors one answer strongly, but they do not measure the actual probability that the classification is correct. We therefore use confidence only to rank cases by uncertainty.

Stage 3: Product category and model reliance. Stage 3 keeps the AI-native decision from Stage 2 unchanged and first assigns each firm to one product category. We designed the categories to distinguish economically different product types rather than group together every firm that mentions AI. Firms classified as not AI-native are separated into traditional technology or software-as-a-service products, AI-augmented products, and non-technology businesses. AI-native firms are classified by where their product sits in the AI value chain and by the principal mechanism through which it creates value: foundation models and infrastructure, tools for AI developers, thin LLM wrappers, thick LLM integrators, applied vertical AI, autonomous agents, or generative-content platforms. The categories describe what the firm sells; they do not, by themselves, indicate how heavily the firm depends on outside models. Table C.3 defines the ten categories.

Table C.3
Product categories used by the AI-native classifier

Product category	Meaning
0A Traditional technology/SaaS	A conventional technology product for which AI is not central to the product's design or function.
0B AI-augmented	AI is a supporting feature rather than the main source of the product's value or functionality.
0C Non-technology	The firm's core offering is not primarily technological.
1A Foundation layer	Core models, model infrastructure, or data layers.
1B AI-native infrastructure and tooling	Tools for building, deploying, monitoring, or governing AI systems.
1C Thin LLM wrapper	A narrow interface whose core functionality depends primarily on third-party LLM APIs.
1D Thick LLM integrator	A foundation model combined with substantial proprietary workflow, data, or product layers.
1E Applied vertical AI	An AI- or machine-learning-based product designed for a specific industry or workflow.
1F Autonomous agent systems	Systems that execute multi-step tasks with limited human intervention.
1G Generative content platform	A platform whose customer deliverable is generated media such as images, audio, video, music, or three-dimensional assets.

For firms classified as AI-native, Stage 3 then measures *Resource-Adjusted AI Dependency* (RAD). We created RAD as a separate dimension because firms in the same product category can differ substantially in their dependence on third-party foundation models and APIs. The organizing question is whether, given substantial funding and five years, the firm could build a competitive version of its product without relying on current third-party foundation-model APIs. The assessment considers evidence of proprietary models, data, fine-tuning, technical scale, deep workflow assets, and the ability to switch model providers; funding or headcount alone does not establish independence.

RAD-H denotes high structural dependency: the product remains tied to outside models or APIs that the firm could not credibly reproduce. RAD-M denotes medium or transitional dependency: the firm uses outside models but has meaningful proprietary capabilities that provide partial independence. RAD-L denotes low structural dependency: the firm develops proprietary AI at its core or has sufficiently independent capabilities for the product to survive the loss of third-party model APIs. Firms classified as not AI-native receive RAD-NA because this dimension does not apply to them.

The model returns separate one-to-five confidence ratings for the product category and RAD decisions. The saved output includes the AI-native indicator, product category, RAD category, both confidence ratings, a short explanation and self-check, cited sources, whether the evidence is current

or archived, and the website snapshot date. The Stage 3 classifier combines a shared classification prompt with the appropriate AI-native or non-AI-native instructions, based on the fixed Stage 2 decision. These three files are listed as Stage 3 in Table C.4.

For transparency, Table C.4 brings together the prompt files used for the LLM-based decisions in Stages 2 and 3. The table lists the files but does not reproduce their contents.

Table C.4
Prompt files used in the AI-native classification pipeline

Stage	Prompt role	File
Stage 2	Binary AI-native classifier	`binary_gate_prompt.txt`
Stage 3	Product-category and RAD classifier	`subclass_rad_prompt.txt`
Stage 3	AI-native category and RAD instructions	`family_block_ai_native.txt`
Stage 3	Non-AI-native category instructions	`family_block_not_ai_native.txt`

Note: Stage 1 is implemented in code and therefore has no LLM prompt file. In Stage 3, `subclass_rad_prompt.txt` is combined with either the AI-native or non-AI-native instruction file, according to the fixed Stage 2 decision.

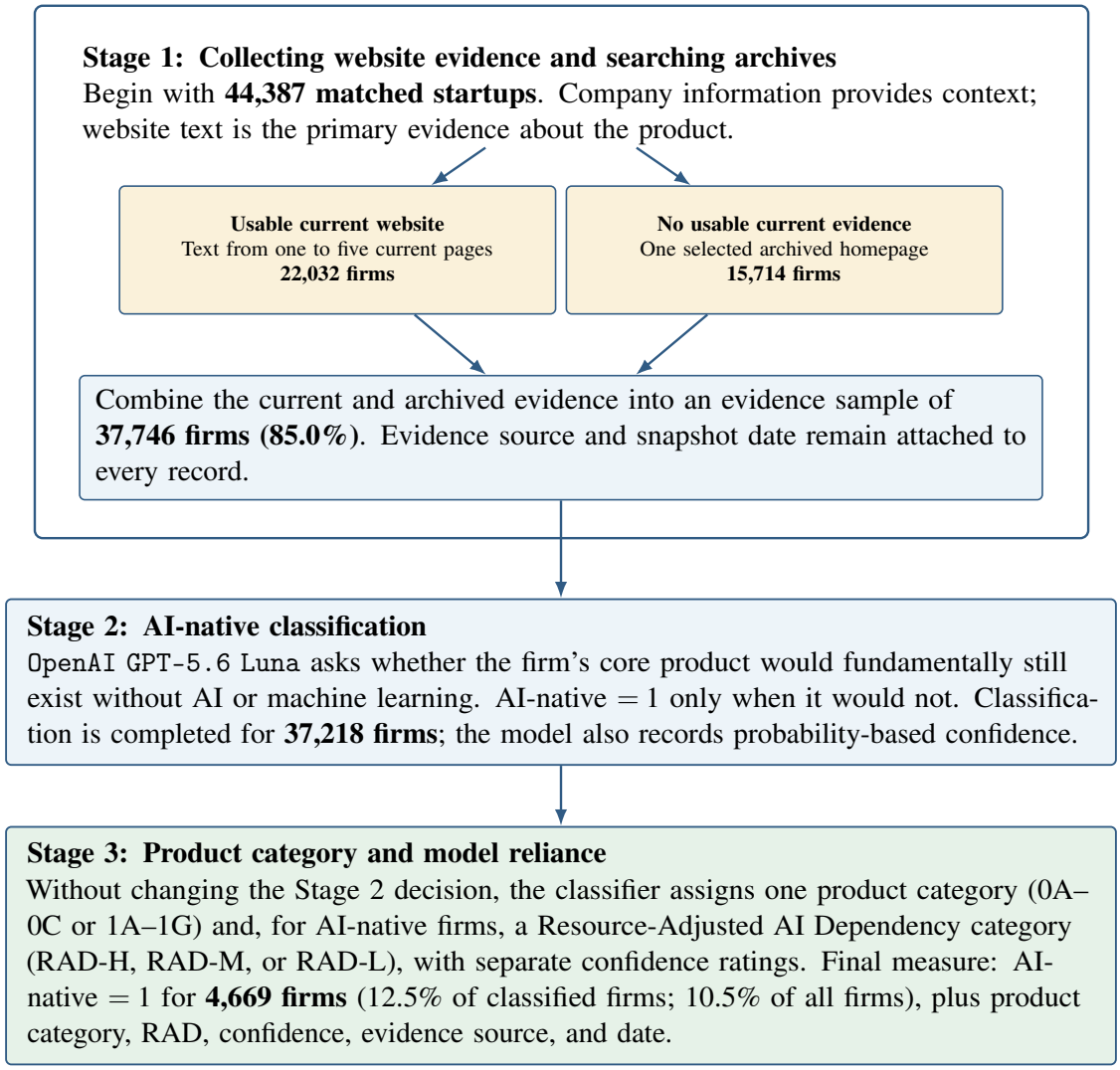


Figure C.2
Workflow for identifying AI-native startups. Searching archived websites expands coverage beyond firms with retrievable current websites. The same classification procedure is applied to current and archived evidence, although the archived source contains less website text.

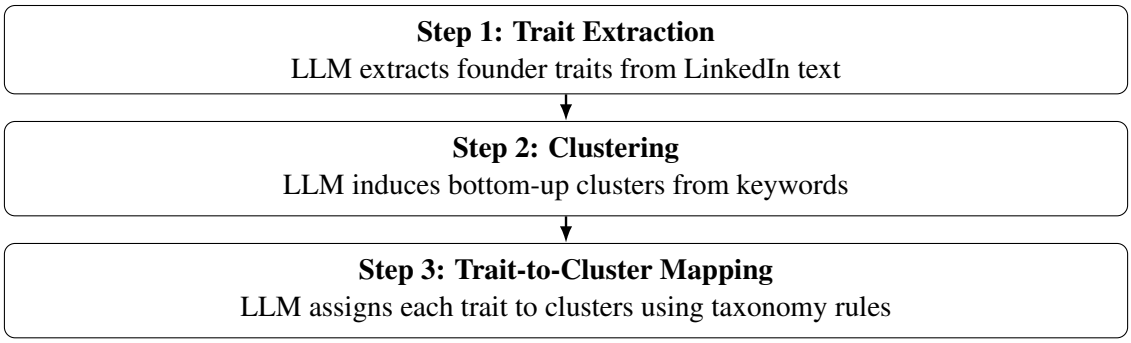
Descriptive patterns. The pipeline classifies 4,669 of the 37,218 firms that completed Stage 2 as AI-native (12.5%), equivalent to 10.5% of the full 44,387-firm sample. The AI-native share is 14.1% among firms classified from current evidence and 10.5% among firms classified from archived evidence. Thus, using only currently retrievable sites would overstate the AI-native share. Using the paper's cohort definition, the share is 9.4% among the 32,231 firms founded in 2020–2022 and 32.7% among the 4,987 firms founded in 2023–2024.

Among the 4,669 AI-native firms, applied vertical AI is the largest product category (59.0%), followed by thick LLM integrators (13.0%) and autonomous agent systems (12.7%); each remaining category accounts for less than 6%. RAD-M accounts for 80.8% of AI-native firms, RAD-H for 10.6%, and RAD-L for 8.6%.

Joint interpretation. The two pipelines do not have identical coverage. The AI-enabled measure adds independent source verification within the fixed 9,420-firm research sample and does not include the Internet Archive search used for AI-native firms. We therefore compare the two indicators only among firms examined by both pipelines. There are 8,252 such firms: 54.3% are AI-enabled, 34.1% are AI-native, and 21.1% satisfy both definitions. This overlap confirms that the two labels capture related but distinct aspects of internal GenAI use and AI-based product design.

D LLM-Assisted Founder Characteristics Comparison

We use a three-step LLM-assisted pipeline to compare founder characteristics for companies established pre- versus post-GenAI.



Step 1: Trait extraction. After matching founders from Crunchbase to LinkedIn profiles, we collect free-text entries that describe each entrepreneur, including profile headline/title, summary ("About"), and job descriptions, restricted to the founding year and the year prior.[1] In our sample, 14,062 of 14,149 pre-GenAI founders and 1,857 of 1,869 post-GenAI founders have non-empty text. We then prompt an LLM to extract concise traits (keywords/attributes) that characterize the person. We also require the LLM to differentiate whether a trait is directly observed or logically inferred; in either case, it also needs to provide the text quote based on which it extracted the trait. This yields over 55k (pre-GenAI) and 11k (post-GenAI) unique traits. The model used in this step is Gemini-3-Flash.

Step 2: Bottom-up cluster induction. We pool traits across both periods ($\sim$ 61k unique traits) and ask an LLM to induce a taxonomy in a strictly bottom-up manner: clusters are formed solely from semantic similarity within the observed trait list, without imposing an external framework. We provide a sequence of tasks for the LLM to follow:

- Normalize keywords by standardizing casing, punctuation, plurals, and hyphenation, and recording near-duplicate aliases.
- Induce bottom-up clusters from the terms present with minimal overlap, marking any necessary multi-membership as multi-label candidate and keeping Misc/Other small.
- Consolidate into a stable final cluster set by merging redundancies and splitting only when clear subgroups exist.

[1] We use LinkedIn profiles observed by the end of 2024. The profile text may therefore include additions or revisions made after the founding year.

- Generate mapping rules for each cluster using strong inclusion cues and clear exclusion cues to avoid nearest-neighbor confusion.
- Add post-hoc academic annotations by attaching relevant organizational behavior or entrepreneurship lenses with citations when confident, or search terms when not.

The model used is GPT-5.2 (thinking mode), and it produces 18 clusters in this unsupervised step.

Step 3: Trait-to-cluster mapping. Finally, we prompt an LLM to assign each trait to 0-3 cluster labels using lexical anchors and boundary rules derived from step 2. Multi-label assignments are allowed when a trait clearly spans multiple domains (e.g., "AI governance" maps to both AI/ML and compliance/security). The model used in this step is GPT-5-mini.

Table D.1
Cluster Description and Representative Examples (1/2)

ID	Cluster name	Inclusion rules (high-signal cues)	Examples
1	Venture roles & startup-stage markers	founder/co-founder; serial entrepreneur; startup CEO/CTO; first employee; accelerator/incubator as role-stage identity	"Co-Founder", "CEO", "Chief Product Officer"
2	Outcomes, traction & scale metrics	ARR/MRR/AUM; revenue/retention/subscribers; NPS; uptime; exit/acquired as outcomes; large numeric/currency growth indicators	"High-growth track record", "Successful exit"
3	Capital, investing & deal mechanics	angel/VC/seed; PE; underwriting; financing structures; capital stack; M&A; portfolio/asset management; exchanges as deal mechanics	"VC-backed", "Fundraising capability"
4	Go-to-market (marketing/sales/growth/ partnerships)	GTM; marketing/sales; demand gen; ABM; channels; partnerships/alliances; ecommerce selling; marketplace growth mechanics	"User Growth", "Marketing Leadership"
5	Product building & "0–1" creation	product manager/owner; product development; launch/MVP/build-from-scratch; 0–1 / $0 \rightarrow 1$ phrasing; builder language tied to creation	"Product development", "Full-cycle Product Experience"
6	Software engineering (apps/web/platforms/ frameworks)	developer roles; programming languages; frameworks/toolkits; app/web development; platform/API implementation terms	"C++", "Mobile App Developer", "Web frontend developer"
7	Data/analytics & experimentation	analytics/BI; dashboards; measurement/attribution; A/B testing; experiment design; insights; regression/statistics phrasing	"Data-driven", "Data Analytics Expert"
8	AI/ML & GenAI (agents, safety, governance)	AI/ML; LLMs; deep learning; computer vision; agents; fine-tuning; AI safety/governance explicitly	"AI/ML focus", "Full-stack AI Developer"
9	Cloud/infra/DevOps & reliability/observability	cloud providers; migrations; infra; observability/monitoring; scalability/reliability; DevOps/SRE terms	"Cloud Infrastructure Developer", "DevOps Specialist"

Table D.2
Cluster Description and Representative Examples (2/2)

ID	Cluster name	Inclusion rules (high-signal cues)	Examples
10	Security, risk, privacy & compliance	security/privacy; risk management; audit/compliance; fraud/AML; governance-as-control/assurance; enforcement	"Compliance Specialist", "Cybersecurity Focus"
11	Design/creative/content production	3D modeling/rendering; animation; Adobe suite; media production; visual/creative craft terms	"Copywriter", "UX Designer", "User-Centric Design"
12	Operations, service delivery & logistics	operations; process; implementation; service delivery; fulfillment/3PL; support systems; logistics	"Inventory Management", "Operational Efficiency", "Process Optimizer"
13	Academic/research/ clinical orientation	professor/researcher; PhD; clinical practice; laboratory; university affiliation as role identity	"Principal Scientist", "Professor", "Researcher"
14	Credentials, memberships, awards & public recognition	certified/certification; fellow/member; awards/honors; rankings; patents as recognition; alumni as credential	"30 Under 30 Honoree", "EY Entrepreneur of the Year"
15	Values, advocacy & impact orientation	advocacy; accessibility; equity/inclusion; affordability; sustainability; ethical/mission-driven for-good language	"Sustainability Focus", "Mission-driven"
16	Personal dispositions & working style	trait adjectives (analytical, adaptable, ambitious, accountable, action-oriented, etc.)	"Client-Centric", "Tech savvy"
17	Industry/vertical domain tags	explicit industry/vertical tags. Sub-cluster tags: 17a Healthcare & life sciences, 17b Financial services / fintech / insurance, 17c Real estate / proptech / housing, 17d Aerospace / defense / industrial / manufacturing, 17e Energy / climate / cleantech, 17f Agriculture / food / agtech, 17g Media / entertainment / gaming / creator economy	"Biomedical expertise", "Commodities trader", "Clean energy focus", "Licensed food manufacturer"
18	Entities/noise/ unclassifiable tokens	opaque proper nouns; very short tokens; numeric-only items; insufficient semantics to match other cues	